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E9
0.992

Nitrosospira

m
ultiformis

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strain

ATCC
25196 / N

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71_

Am

m

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niu

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sporter Q2YB

P7

0.932
0.767

Nitrosospira

m
ultiformis

Am

m

o

niu

m tra

n

sporter A0A1I0AQM4

0.9

Nitrosospira

m
ultiformis

Am

m

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niu

m tra

n

sporter A0A2T5IH

P0

Nitrosospira

m
ultiformis

Am

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sporter A0A1I7

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Nitroso

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ATCC 19718 / CIP 103999 / KCTC

2705 / N

B

RC 14298_

Am

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niu

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Nitrosospira

m
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Am

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nsporter A0A1H8D0R8

0.616

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Am

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niu

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nsporter AmtB-like

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ntaining

protein
A9V5W4

0.593

0.807

Naegleria

gru
beri Rh-like
glycoprotein

D2V6P8

0.975

Dictyosteliu

m
purpureu

m
Am
moniu

m tra

nsporter AmtB-like
domain-containing

protein
F0ZCX7

0.87

Dictyosteliu

m

discoideu

m

Rhesus-like
glycoprotein

B
Q8I0L6

Dictyosteliu

m

discoideu

m

Rhesus-like
glycoprotein

A
Q9NIV0

0.493

Mus

m
usculus

Am
moniu

m transporter Rh type

A
Q9QUT0

0.452

0.749

0.895

Homo
sapiens

Am
moniu

m transporter Rh type

C

Q9UBD6 1 Mus musculus Ammonium transporter Rh type

C
Q9QXP0

Rattus
norvegicus

Am
moniu

m transporter Rh type

C
Q7TNN9

0.792

0.51

Takifug

u ru
bripes

Am
moniu

m transporter Rh type

C

1
Q1
8PF5

Danio rerio

Am
moniu

m transporter Rh type

C-like

2
Q8JI1

4

0.81

5

Takifugu ru

bripes

Am
moniu

m transporter Rh type

C

2

Q7T3R4

0.6

Kryptolebias
marmoratus
Rhesus
glycoprotein
A6XBM2

nan

0.327

0.79

Danio rerio
Ammoniu

m

transporter Rh

type

B
Q7T070

0.99
Mus
musculus
Ammoniu

m

transporter Rh

type

B

Q8BUX5

Homo
sapiens
Ammonium
transporter Rh

type

B
Q9H31

0

0.362

Eptatretus
stoutii RhcG

A0A1
W5PRD1

0.37

Nematostella
vectensis
Ammonium

transporter AmtB-like
domain-containing

protein
A7SED8

0.362
0.99
Carcinus
maenas
Rh-like
protein
Q95P70

0.98

Metacarcinus

magister
Rh-like
protein
F2YIC4

Portunus
trituberculatus

Rh
protein
A0A0B4M1

63

0.371

Apis
mellifera
Rh-like
protein

A0A7M6UUM4

0.394

0.2

Caenorhabditis

elegans
Ammonium
transporter
AmtB-like

domain-containing

protein
Q22947

Acanthamoeba
castellanii _

strain
ATCC

3001

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Neff_
Ammonium
transporter
subfamily

protein
L8H5K0

0.435

Branchiostoma

floridae
Rh-related

protein
D7UQE2

0.462

0.854

Branchiostoma

floridae
Rh-related

protein
D7UQE4

0.65

Branchiostoma

floridae
Rh-related

protein
D7UQE3

0.66

Takifugu
rubripes
RHAG-1
Q7T3R8

0.66

Ailuropoda
melanoleuca
Ammonium
transporter

Rh
type
C _
Fragment_
D2HCE3

0.52

Oreochromis

grahami Rhag Dani S4TD66 o rerio Ammonium

transporter

Rh
type

A

Q7T067

0.502

Daphnia

pulex

Ammonium

transporter AmtB-like
domain-containing

protein
E9HAT0

0.608

Cryptococcus
neoformans
var. grubii c45

MFS
transporter
A0A225YUR6

0.74

Aspergillus
fumigatus

_

strain
ATCC
MYA-4609

/
CBS
101355

/

FGSC A1100
/ Af293_

Major
facilitator
superfamily
multidrug
transporter

mdr3
Q4WF45

0.595

Neurospora

crassa

_

strain
ATCC
24698

/

74-OR23-1A /

CBS
708.71

/
DSM
1257

/
FGSC
987_
Major
facilitator
superfamily

_

MFS_

profile

domain-containing

protein
Q7S343

0.88

Debaryomyces
hansenii _

strain
ATCC
36239

/
CBS

767

/

BCRC
21394

/
JCM
1990

/

NBRC
0083
/ IGC
2968_

DEHA2C03718p

Q6BVC6

Candida
dubliniensis

_

strain
CD36
/ ATCC
MYA-646

/
CBS
7987

/

NCPF
3949

/

NRRL
Y-17841_

Major
facilitator
superfamily

_

MFS_

profile

domain-containing

protein
B9W978

0.756

0.791

1
Mus
musculus
Solute
carrier
family

22

member

2

O70577

Rattus
norvegicus
Solute
carrier
family

22

member

2

Q9R0W2

0.876
0.823

Dactylococcopsis

salina

_

strain
PCC
8305_
Ammonia
permease
K9Z0D3

0.975

Leisingera
sp. ANG1
Ammonium
transporter
A0A0B4E6X7

1

Lutibaculum
baratangense

AMV1

Ammonium
transporter
V4RHR4

Fulvimarina
pelagi HTCC2506
Ammonium
transporter
family
protein
Q0G2T1

0.719

0.695

Mycolicibacterium
parafortuitum
Putative
ammonium
transporter _
Octadecabacter

arcticus
238_

A0A375YPN6

0.98

Homo
sapiens
Proton

channel OTOP1
Q7RTM1

Danio
rerio
Proton

channel OTOP1
Q7ZWK8

0.703

0.97

Lutibaculum

baratangense AMV1
Ammonium
transporter
V4RK08

Fulvimarina
pelagi HTCC2506
Ammonium
transporter
family
protein
Q0G2V0

0.56

Roseobacter litoralis _

strain
ATCC
49566

/
DSM
6996
/ JCM
21268

/

NBRC
15278
/ OCh
149_
Putative
ammonium
transporter
F7ZJZ1

0.7

Nitrincola lacisaponensis

Ammonium
transporter
A0A063Y394

Marinomonas
mediterranea _

strain
ATCC
700492
/ JCM
21426 /
NBRC
103028 /
MMB-1_
Ammonium
transporter
F2JYR9

0.567

0.803

0.3

Coraliomargarita
akajimensis _

strain
DSM
45221
/ IAM
15411
/ JCM
23193 /
KCTC

12865 / 04OKA010-24_

Ammonium
transporter
D5EP68

Phycisphaera
mikurensis _

strain
NBRC
102666 /
KCTC
22515 /

FYK2301M01_
Ammonium

transporter I0ICG0

0.81

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Fibrobacter
succinogenes _
strain ATCC
19169 / S85_
Ammonium
transporter
C9RS76

Fibrobacter
succinogenes
Ammonium
transporter
A0A380S5X4

0.83

0.28

Plesiocystis
pacifica
SIR-1

Ammonium
transporter
A6GJF5

Phaeodactylibacter
xiamenensis
Ammonium
transporter
A0A098S5Y7

0.861

0.68

Desulfotignum
phosphitoxidans

DSM
13687

Ammonium
transporter
S0G181

Desulfurivibrio
alkaliphilus _

strain
DSM

19089 / UNIQEM
U267 / AHT2_
Ammonium
transporter
D6Z5V9

0.892

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Lentisphaera
araneosa
HTCC2155
Ammonium
transporter
A6DN47

Lentisphaera
araneosa
HTCC2155
Ammonium
transporter
A6DUG4

0.87

0.967

Marivirga
tractuosa _

strain ATCC 23168 / DSM 4126 / NBRC

15989 / NCIMB
1408 / VKM
B-1430 / H-43_
Ammonium
transporter
E4TRS4

0.984

Fulvivirga imtechensis AK7

Ammonium
transporter
L8JUH3

0.953
0.72

Cyclobacterium
marinum _

strain ATCC 25205 / DSM

745 / LMG
13164 / NCIMB

1802_
Ammonium
transporter
G0IYI2

Echinicola
vietnamensis _

strain
DSM

17526 / LMG 23754 / KMM

6221_
Ammonium
transporter
L0FU80

0.995

Rhodonel um
psychrophilum
GCM71 _

DSM
17998

Ammonium
transporter
U5C4L4

0.76

Aquiflexum
balticum

DSM
16537

Ammonium
transporter
A0A1W2H3F0

Mariniradius

saccharolyticus AK6
Ammonium
transporter
M7XJ51

0.655

0.752

Cytophaga
hutchinsoni

_

strain ATCC 33406 / DSM

1761
/ CIP

103989 / NBRC

15051

/ NCIMB 9469 / D465_

Ammonium
transporter
A0A6N4SPH9

0.852
0.63

Imtechel a
halotolerans

K1

Ammonium
transporter
I0WJ97

0.63

Mesoflavibacter zeaxanthinifaciens
subsp. sabulilitoris
Ammonium
transporter
A0A2T1N7R1

Formosa
agariphila _

strain
DSM

15362 / KCTC

12365 / LMG 23005 / KMM 3901

/ M-2Alg
35-1_
Ammonium
transporter
T2KGT3

0.545

Robiginitalea
biformata _

strain ATCC BAA-864 / HTCC2501

/ KCTC
12146_
Ammonium
transporter
A4CMI3

0.5

Psychroflexus gondwanensis ACAM 44

Ammonium
transporter
N1WR17

Arenibacter certesi
Ammonium
transporter
A0A918MJ52

0.924

0.7

Anditalea
andensis
Ammonium
transporter
A0A074KY88

Microscil a

marina ATCC 23134
Ammonium
transporter
A1ZGK6

0.81

Cyclobacterium
marinum _

strain ATCC 25205 / DSM 745 / LMG

13164 / NCIMB

1802_
Ammonium
transporter
G0J614

0.802

Anditalea
andensis
Ammonium
transporter
A0A074KZD8

0.905
0.7

Nitritalea
halalkaliphila

LW7

Ammonium
transporter
I5BYV5

Aquiflexum
balticum
DSM
16537

Ammonium
transporter
A0A1W2GZ82

0.485

Rhodonel um psychrophilum

GCM71 _

DSM
17998

Ammonium
transporter
U5C2K4

0.56

Bel iel a
baltica _

strain
DSM

15883 / CIP 108006 / LMG 21964 / BA134_

Ammonium
transporter
I3Z3H7

Cecembia
lonarensis _

strain

CCUG 58316 / KCTC 22772 / LW9_

Ammonium
transporter
K1LDP0

0.41

0.634

0.592

Acidobacterium
capsulatum _

strain ATCC 51196 / DSM 11244 / BCRC 80197 / JCM 7670 / NBRC 15755 / NCIMB 13165 / 161_

Ammonium
transporter
C1F584

0.603

0.757
0.66

Haliangium
ochraceum _

strain

DSM 14365 / JCM 11303 / SMP-2_

Ammonium
transporter
D0LTW3

1

Anaeromyxobacter dehalogenans _

strain

2CP-1 / ATCC BAA-258_

Ammonium
transporter
B8J5X4

Anaeromyxobacter dehalogenans _

strain

2CP-C_
Ammonium
transporter
Q2IK04

0.834

1

Leptonema

il ini

Ammonium transporter
A0A833LX85

Leptonema

il ini DSM 21528

Ammonium transporter

H2CLF7

0.894

Leptospira

biflexa serovar Patoc _

strain
Patoc

1 / ATCC 23582 / Paris_
Ammonium transporter

B0SL05

0.911

Arcobacter nitrofigilis _

strain ATCC 33309 / DSM 7299 / CCUG 15893 / LMG 7604 / NCTC 12251 / CI_

Ammonium transporter

D5V7B3

0.937

0.757

Acaryochloris
marina _

strain

MBIC 11017_

Ammonium transporter

B0BZN3

0.76

Microcystis aeruginosa

NIES-3804 Ammonium transporter

A0A6H9GLV4

0.64

Microcystis aeruginosa

NIES-3787 Ammonium transporter

A0A6H9FPY5

0.48

Crocosphaera watsoni

WH 0005 Ammonium transporter

T2J0N1

Moorena producens 3L Ammonium transporter

F4Y3C5

0.883

0.779

1

Cylindrospermopsis raciborski
CS-505 Ammonium transporter

A0A853MG28

Cylindrospermopsis raciborski
CENA303 Ammonium transporter

A0A1X4GBG2

0.894

Chroococcidiopsis thermalis _

strain

PCC 7203_

Ammonium transporter

K9U8C7

0.855

Crinalium epipsammum

PCC 9333 Ammonium transporter

K9W4V5

0.86

Fischerel a muscicola CCMEE 5323 Ammonium transporter

A0A2N6JV02

0.995

Nodularia spumigena CENA596 Ammonium transporter

A0A166IS52

1

Nodularia spumigena CCY9414 Ammonium transporter

A0ZJU7

Nodularia spumigena UHCC 0039 Ammonium transporter

A0A2S0Q8P1

0.836

0.98

Microcystis aeruginosa NIES-3787 Ammonium transporter

A0A6H9GEE4

Microcystis aeruginosa NIES-3804 Ammonium transporter

A0A6H9GWG4

0.696
0.783

Cyanobacterium stanieri _
strain ATCC 29140 / PCC 7202_
Ammonium transporter

K9YN27

0.995

Planktothrix agardhi

CCAP 1459/11A Ammonium transporter

A0A4V0XUI3

0.7

Planktothrix agardhi
Ammonium transporter
A0A1J1JH03

Planktothrix agardhi

NIVA-CYA 126/8 Ammonium transporter A0A073CP02

0.538

Moorena producens 3L Ammonium transporter

F4XTK4

0.573

1

Crocosphaera watsoni WH 8502 Ammonium transporter

T2IEJ2

Crocosphaera watsoni WH 8501
Ammonium transporter

Q4CA23

0.4
5

Cyanobacterium stanieri _
strain ATCC 29140 / PCC 7202_
Ammonium transporter

K9YJQ3

0.4

Rubidibacter lacunae KORDI 51-2 Ammonium transporter

U5DEY6

Dactylococcopsis salina _
strain PCC 8305_
Ammonium transporter

K9YR86

0.934

Prochlorococcus marinus _
strain MIT 9303_

Ammonium transporter A2CCK1

0.958

Cyanobium gracile _

strain ATCC 27147 / PCC 6307_
Ammonium transporter

K9P718

0.9 2

Prochlorococcus marinus _
strain MIT 9313_

Ammonium transporter

Q7V4T8

0.949

Prochlorococcus marinus str. MIT 91
6 Ammonium transporter A0A0A1ZMZ9

0.8 7

1

Prochlorococcus marinus XMU1408 Ammonium transporter A0A318QZ80

1

Prochlorococcus marinus _
strain NATL1A_

Ammonium transporter A2C095

0.53

Prochlorococcus marinus str. PAC1 Ammonium transporter A0A0A2C6L1

Prochlorococcus marinus _
strain NATL2A_

Ammonium transporter

Q46HB1

0.81

Prochlorococcus marinus str. MIT 9302 Ammonium transporter A0A0A2A5X1

0.805
0.5

Prochlorococcus marinus _
strain MIT 9312_

Ammonium transporter

Q31CR9

Prochlorococcus marinus _
strain MIT 9515_

Ammonium transporter A2BUP4

0.897

Prochlorococcus marinus Ammonium transporter

Q9REX3

0.93

Prochlorococcus marinus _
strain MIT 9215_

Ammonium transporter A8G2S3

0.89

0.63

Prochlorococcus marinus str. MIT 9201 Ammonium transporter A0A0A2A0Y9

Prochlorococcus marinus str. GP2 Ammonium transporter A0A0A1ZJU5

0.825

0.69

Prochlorococ

us marinus str. MIT 9314 Ammonium transporter A0A0A2AMW2

Prochlorococ
us marinus _
strain MIT 9301_

Ammonium transporter A3PAY4

0.63

Prochlorococ

us marinus str. MIT 9401 Ammonium transporter A0A0A2BBY0

0.57

Prochlorococ
us marinus _
strain AS9601_

Ammonium transporter A2BP62

Prochlorococ us marinus str. SB Ammonium transporter A0A0A2B651

0.97

Desulfurispiril um indicum _

strain ATCC BA -1389 / DSM 2 839 / S5_
Ammonium transporter

E6W5C1

0.976

0.763

0.652

0.96

Geoalkalibacter fer ihydriticus Ammonium transporter A0A1G9P6Q0 Geoalkalibacter subter aneus Ammonium transporter A0A0B5FNH4

0.78

1

Oceanibaculum indicum P24 Ammonium transporter

K2JVF8
Oceanibaculum indicum Ammonium transporter A0A420WC79

1

Desulfuromonas acetoxidans _
strain DSM 684 / 1 070_

Ammonium transporter Q1K0H9

Desulfuromusa kysingi Ammonium transporter A0A1H4DJ49

0.756

0.453

0.54

Hyphomonas neptunium _
strain ATCC 154

_

Ammonium transporter family protein Q0BWD0

Hirschia baltica _

strain ATCC 49814 / DSM 5838 / IFAM 1418_
Ammonium transporter C6XKC2

0.08

Parvularcula bermudensis _

strain ATCC BA -594 / HTCC2503 / KCTC 12087_

Ammonium transporter

E0TGP9

Pannonibacter phragmitetus Ammonia transporter A0A379A0S9

0.732

0.705

Parvularcula bermudensis _

strain ATCC BA -594 / HTCC2503 / KCTC 12087_
Ammonium transporter E0TCV4

0.9

Maricaulis maris Ammonium transporter A0A495D4X4 Maricaulis maris _strain MCS10_ Ammonium transporter Q0AT01

0.78

0.687
0.54

Fulvimarina pelagi HTCC2506 Ammonium transporter Q0G0A3

Aurantimonas manganoxydans _

strain ATCC BA -12 9 / DSM 21871 / SI85-9A1_
Ammonium transporter Q1YDY8

1

Nitratireductor aquibiodomus Ammonium transporter A0A1H4KAQ7 Nitratireductor aquibiodomus RA2 Ammonium transporter I5C04

0.7

0.56

Altererythrobacter ishigakiensis Ammonium transporter A0A562USB9

0.625

0.98

Martelel a mediter anea DSM 17316 Ammonium transporter A0A1U9Y 82 Martelel a mediter anea Ammonium transporter A0A4R3NVT3

0.58

Lutibaculum baratangense AMV1 Ammonium transporter V4REY8

0.45

Rhodospiril um rubrum _strain ATCC 1

70 / ATH 1.1.1 / DSM 467 / LMG 4362 / NCIMB 825 / S1_ Rh-like protein/ammonium transporter Q2RWY7

Bauldia litoralis Ammonium transporter A0A1G6ABX1

0.93

0.865

Leisingera sp. ANG1 Ammonium transporter A0A0B4DUR9

0.84

Actibacterium mucosum KCTC 23 49 Ammonium transporter A0A037ZMW1 Pelagibacterium halotolerans _strain DSM 2 347 / JCM 157 5 / CGMCC 1.7692 / B2_ Ammonium transporter G4R7P1

0.838

0.9 7

0.4

Dinoroseobacter shibae _strain DSM 16493 / NCIMB 14021 / DFL 12_ Ammonium transporter A8LKQ0

Pseudo ceanicola batsensis _strain ATCC BAA-863 / DSM 15984 / KCTC 12145 / HTCC2597_ Ammonium transporter A3TVQ9

0.64

Roseobacter litoralis _strain ATCC 4956 / DSM 69 6 / JCM 21268 / NBRC 15278 / OCh 149_ Ammonium transporter F7ZEU9 Octadecabacter arcticus 238 Ammonium transporter M9RKT6

0.85

Phaeobacter gal aeciensis Ammonium transporter A0A1B0ZRL8

0.94

Nereida ignava Ammonium transporter A0A0U1NJW3

Phaeobacter gal aeciensis Ammonium transporter A0A36 WYI9

0.9 2

Marichromatium purpuratum 984 Ammonium transporter W0DYY3

0.978

0.583

Alkali mnicola ehrlichi Ammonium transporter A0A3E0X01

0.62

Gal aecimonas xiamenensis 3-C-1 Ammonium transporter K2IMJ7

0.675

Ectothiorhodospira haloalkaliphila Ammonium transporter W8KM07

0.45

Halorhodospira halochloris Ammonium transporter A0A0X8X7Y7

Alkali mnicola ehrlichi _strain ATCC BAA-1 01 / DSM 17681 / MLHE-1_ Ammonium transporter Q0ACK1

0.895

Pseudomonas berkeleyensis Ammonium transporter A0A7G5DPV1

0.8

0.59

Nitrincola lacisaponensis Ammonium transporter A0A063Y780

0.6 3

1

Halomonas elongata Ammonium transporter A0A329Q0D7
Halomonas elongata _strain ATCC 3 173 / DSM 2581 / NBRC 15 36 / NCIMB 2198 / 1H9_ Ammonium transporter E1V4T3

0.8

Chromohalobacter salexigens _strain ATCC BAA-138 / DSM 3043 / CIP 106854 / NCIMB 13768 / 1H1 _ Ammonium transporter Q1QSQ2 Halotalea alkali enta Ammonium transporter A0A172YBR4

0.856

0.5 6

Endozoicomonas montiporae CL-3 Ammonium transporter A0A142BI19

0.525

Desulfonatronospira thiodismutans ASO3-1 Ammonium transporter D6SPD1

0.4 7

1

Desulfocapsa sulfexigens _strain DSM 10523 / SB164P1_ Ammonium transporter M1P5F0

Desulfocapsa sulfexigens _strain DSM 10523 / SB164P1_ Ammonium transporter M1NZT5

0.29

Desulfotalea psychrophila _strain LSv54 / DSM 12343_ Ammonium transporter Q6ARH0 Idiomarina loihiensis _strain ATCC BAA-735 / DSM 15497 / L2-TR_ Ammonia permease Q5QVH2

0.8 9

Fer imonas balearica _strain DSM 979 / CCM 4581 / KCTC 23876 / PAT_ Ammonium transporter E1SLM7

0.893

Congregibacter litoralis KT71 Ammonium transporter A4ABK3

0.872

0.4

Cel vibrio japonicus _strain Ueda107_ Ammonium transporter B3PF7

0.73

Alishewanel a agri BL06 Ammonium transporter I9P0 3

Rheinheimera nanhaiensis E407-8 Ammonium transporter I1DSU5

0.858
0.29

Salinisphaera shabanensis E1L3A Ammonium transporter protein U2EN72

Hahel a chejuensis _strain KCTC 2396_ Ammonium transporter Q2S8H9

0.876

Alcanivorax borkumensis _strain ATCC 70 651 / DSM 1 573 / NCIMB 13689 / SK2_ Ammonium transporter Q0VLH1

0.862

Reinekea blandensis MED297 Ammonium transporter A4BAE1

0.869

Neptuni bacter caesariensis Putative ammonium transporter A0A7U8C74

0.861

Sac harophagus degradans _strain 2-40 / ATCC 43961 / DSM 17024_ Rh-like protein/ammonium transporter Q21P82

0.827

1

Alteromonas macleodi Ammonium transporter AmtB-like domain-containing protein A0A6T9XUR2

0.8

Alteromonas mediter anea 615 Ammonium transporter S5ABR8
Alteromonas mediter anea _strain DSM 171 7 / CIP 1 0805 / LMG 28347 / De p ecotype_ Ammonium transporter F2G6G4

0.754

0.07

Catenovulum agarivorans DS-2 Ammonium transporter W7QG13

0.03

Agarivorans albus MKT 106 Ammonium transporter R9PI53 Psychrobacter cryohalolentis _strain ATCC BAA-12 6 / DSM 17306 / VKM B-2378 / K5_ Ammonium transporter Q1Q879

0.642

Pseudoalteromonas atlantica _strain T6c / ATCC BAA-1087_ Ammonium transporter A0A9J9BAE8

0.651
0.475

Hydrogenovibrio marinus Ammonium transporter A0A06 ZTE4

0.353

0.2

Flexistipes sinusarabici _strain ATCC 49648 / DSM 4947 / MAS 10_ Ammonium transporter F8E6R1

Methylophaga frap ieri _strain ATCC BAA-2434 / DSM 25690 / JAM7_ Ammonium transporter I1YJH7

0.4

Marinomonas mediter anea _strain ATCC 70 492 / JCM 21426 / NBRC 103028 / MMB-1_ Ammonium transporter F2JTN3 Neptunomonas japonica JAMM 1380 Ammonium transporter A0A7R6PER3

0.81

0.9

Colwel ia psychrerythraea Ammonium transporter A0A1Y5E8I7
Colwel ia psychrerythraea _strain 34H / ATCC BAA-681_ Putative ammonium transporter Q48 X2

0.96

Moritel a marina ATCC 15381 Ammonium transporter A0A5J6WQU1

0.89

Psychromonas ingrahami _strain DSM 176 4 / CCUG 5185 / 37_ Ammonium transporter A1SZ12

0.983

0.9

Enterovibrio norvegicus FF-454 Ammonium transporter A0A1E5BXA4 Enterovibrio norvegicus Ammonium transporter A0A2N7LCD7

0.94

Photobacterium profundum 3TCK Ammonium transporter Q1Z4I4 Photobacterium profundum _strain SS9_ Ammonium transporter Q6LMI6

0.866

0.363

0.572

0.582

Rhodothermus marinus _strain ATCC 43812 / DSM 4252 / R-10_ Ammonium transporter D0MFE8

0.723

Methanofol is liminatans DSM 4140 Ammonium transporter J0S2I0

0.95

Niastel a kore nsis _strain DSM 17620 / KACC 1 465 / NBRC 106392 / GR20-10_ Ammonium transporter G8TK79

0.71

Fimbri monas ginsengisoli Gsoil 348 Ammonium transporter A0A068NV61

0.49

Ktedonobacter racemifer DSM 4 963 Ammonium transporter D6TFE1 Mucilaginibacter paludis DSM 18603 Ammonium transporter H1YF74

0.616

0.98

Roseiflexus castenholzi _strain DSM 13941 / HLO8_ Ammonium transporter A7NKH2

Chloroflexus aurantiacus _strain ATCC 29366 / DSM 635 / J-10-fl_ Ammonium transporter A9WHW6

0.622

0.766

0.278
0.62

Monosiga brevicol is Ammonium transporter A9UR54

0.88

Micromonas pusil a _strain CCMP1545_ Ammonium transporter C1N6X9

0.34

Micromonas pusil a Ammonium transporter A0A7S0KKL8 Micromonas pusil a _strain CCMP1545_ Ammonium transporter C1N5A7

0.292

0.77

Perkinsus marinus _strain ATCC 50983 / TXsc_ Ammonium transporter C5LAG4

Perkinsus marinus _strain ATCC 50983 / TXsc_ Ammonium transporter C5KTM6

0.39
0.02

Cyanidioschyzon merolae _strain NIES-3377 / 10D_ Probable ammonium transporter M1V7W3

Acanthamoeba castel ani _strain ATCC 30010 / Nef _ Ammonium transporter L8H844

1

Emiliania huxleyi Ammonium transporter R1EDV9

Emiliania huxleyi Ammonium transporter R1FI09

0.475

0.552
0.69

Aureococ us anophagef erens Putative ammonium transporter F0Y9X3

0.985

Aureococ us anophagef erens Uncharacterized protein AMT4 F0YLV7

1

Emiliania huxleyi Ammonium transporter R1EGK6

Emiliania huxleyi Ammonium transporter R1D1H4

0.797
0.14

Micromonas pusil a Ammonium transporter A0A6U2BIY7

Ectocarpus siliculosus Ammonium transporter D7G265

0.998

1

0.68

Oxyr his marina Ammonium transporter A0A7S4GNJ2

Oxyr his marina Ammonium transporter A0A7S4GNF3

0.53

Oxyr his marina Ammonium transporter A0A7S4GNI0

Oxyr his marina Ammonium transporter A0A7S4GND2

1

Alexandrium tamarense Ammonium transporter A0A0K2STH0

0.99

Alexandrium tamarense Ammonium transporter A0A0K2STK6

Alexandrium tamarense Ammonium transporter A0A0K2STQ7

0.472

0.735

Chlamydomonas reinhardti Ammonium transporter Q6QH01

0.95

Chlamydomonas reinhardti Ammonium transporter A0A2K3DQF8

Chlamydomonas reinhardti Ammonium transporter A0A2K3DWA2

0.556

Guil ardia theta _strain CCMP2712_ Ammonium transporter L1INN6

0.624

0.55

Solanum lycopersicum Ammonium transporter 1 member 2 O04161

Naegleria gruberi Ammonium transporter D2V6Q5

0.776

0.98

Phaeodactylum tricornutum _strain CCAP 1055/1_ Ammonium transporter B7G0Y4

Oxyr his marina Ammonium transporter A0A7S4GMD8

0.78

0.87

Phaeodactylum tricornutum _strain CCAP 1055/1_ Ammonium transporter _Fragment_ B7FW68

Phaeodactylum tricornutum _strain CCAP 1055/1_ Ammonium transporter _Fragment_ B7G0Y1

0.89

Phaeodactylum tricornutum _strain CCAP 1055/1_ Ammonium transporter _Fragment_ B7FSA7

Phaeodactylum tricornutum _strain CCAP 1055/1_ Ammonium transporter _Fragment_ B7FTM9

0.893

0.825

Halogranum salarium B-1 Ammonium transporter J2ZGN6

0.769

0.981
0.65
0.8

Halopiger xanaduensis _strain DSM 18323 / JCM 14033 / SH-6_ Ammonium transporter F8D7H6

Natrinema pellirubrum _strain DSM 15624 / CIP 106293 / JCM 10476 / NCIMB 786 / 157_ Ammonia permease L0JT30

0.28

Natronorubrum aibiense Ammonium transporter A0A5P9P1L2
Haloter igena turkmenica _strain ATCC 51198 / DSM 5511 / JCM 9101 / NCIMB 13204 / VKM B-1734 / 4k_ Ammonium transporter D2RU19

0.595

Halostagnicola larsenii XH-48 Ammonium transporter W0JIB3

0.51
0.47

Halorhabdus utahensis _strain DSM 12940 / JCM 11049 / AX-2_ Rh family protein/ammonium transporter C7NND3

Halosimplex carlsbadense 2-9-1 Ammonium transporter M0D3Q9

0.74

Halomicrobium mukohataei _strain ATCC 700874 / DSM 12286 / JCM 9738 / NCIMB 13541_ Ammonium transporter C7P0S0 Haloarcula marismortui _strain ATCC 43049 / DSM 3752 / JCM 8966 / VKM B-1809_ Ammonium transporter Q5UZ98

0.508

Haloferax mediter anei _strain ATCC 33500 / DSM 1411 / JCM 8866 / NBRC 14739 / NCIMB 2177 / R-4_ Ammonium transporter Amt2 B8ZYW3

0.486

Halalkalicoc us jeotgali _strain DSM 18796 / CECT 7217 / JCM 14584 / KCTC 4019 / B3_ Rh family protein/ammonium transporter D8J4C6

0.583

Halorubrum lacusprofundi _strain ATCC 49239 / DSM 5036 / JCM 8891 / ACAM 34_ Rh family protein/ammonium transporter B9LU00

0.693

Halococ us sac harolyticus DSM 5350 Rh family protein/ammonium transporter M0MDP9 0.975
Methanococ us maripaludis X1 Ammonium transporter G0H0Z6

0.73

Methanotor is igneus _strain DSM 5666 / JCM 11834 / Kol 5_ Ammonium transporter F6BEI0

Methanococ us maripaludis _strain C5 / ATCC BAA-1333_ Ammonium transporter A4G0C5

0.973

Halosimplex carlsbadense 2-9-1 Ammonium transport protein M0D580

0.909

Halogeometricum borinquense Ammonium transporter A0A482TED1

0.822

Halogranum salarium B-1 Ammonium transporter J3EXK0

0.806

Haloferax mediter anei _strain ATCC 33500 / DSM 1411 / JCM 8866 / NBRC 14739 / NCIMB 2177 / R-4_ Ammonium transporter M0J579

0.868

Haloferax mediterranei _strain ATCC 33500 / DSM 1411 / JCM 8866 / NBRC 14739 / NCIMB 2177 / R-4_ Ammonium transporter Amt1 I3R0S7 0.98 Haloquadratum walsbyi J07HQW2 Ammonia permease U1PPI7

0.68

Haloquadratum walsbyi J07HQW1 Ammonia permease U1PIR6

1

Haloquadratum walsbyi _strain DSM 16790 / HBSQ001_ Transport protein _Probable substrate ammonium_ Q18ES8

Haloquadratum walsbyi _strain DSM 16854 / JCM 12705 / C23_ Transport protein _Probable substrate ammonium_ G0LFE3

0.277

0.491

0.773

Alcanivorax borkumensis _strain ATCC 700651 / DSM 11573 / NCIMB 13689 / SK2_ Ammonium transporter Q0VPH4

0.455

Leptothrix cholodnii _strain ATCC 51168 / LMG 8142 / SP-6_ Ammonium transporter B1XZU2

0.51

Methylobacillus flagellatus _strain KT / ATCC 51484 / DSM 6875_ Ammonium transporter Q1H0G2

Pseudomonas berkeleyensis Ammonium transporter A0A7G5DKA9

0.524

0.523

1

Kuenenia stut gartiensis Ammonium transporter A0A2C9CK98

Kuenenia stut gartiensis Ammonium transporter Q1Q357

0.9

Desulfosporosinus orientis _strain ATCC 19365 / DSM 765 / NCIMB 8382 / VKM B-1628 / Singapore I_ Ammonium transporter G7WFX2

Desulfosporosinus acidiphilus _strain DSM 22704 / JCM 16185 / SJ4_ Ammonium transporter I4D6Z9

0.588

Marichromatium purpuratum 984 Ammonium transporter W0E077

0.598

0.383

0.2

Catenovulum agarivorans DS-2 Diguanylate cyclase/phosphodiesterase _GGDEF & EAL domains_ with PAS/PAC sensor_S_ W7QGM4

Hydrogenovibrio marinus Ammonium transporter A0A066ZRQ5

1

Leptonema illini Ammonium transporter A0A833GZR2

Leptonema illini DSM 21528 Ammonium transporter H2CJI0

0.665

0.36

0.36

Psychromonas ingrahamii _strain DSM 17664 / CCUG 51855 / 37_ Ammonium transporter A1SY01

Glaciecola nitratireducens _strain JCM 12485 / KCTC 12276 / FR1064_ Diguanylate cyclase/phosphodiesterase with PAS/PAC and GAF sensor_S_ G4QHD2

0.62

Acaryochloris marina _strain MBIC 11017_ Ammonium transporter A8ZQD5

Desulfomicrobium baculatum _strain DSM 4028 / VKM B-1378 / X_ Diguanylate cyclase/phosphodiesterase with PAS/PAC sensor_S_ C7LQR9

0.792

0.06 Roseobacter litoralis _strain ATCC 49566 / DSM 6996 / JCM 21268 / NBRC 15278 / OCh 149_ Ammonium transporter F7ZJM0
Apis mellifera Amt-2-like protein A0A7M6UML9

0.847 Parvibaculum lavamentivorans _strain DS-1 / DSM 13023 / NCIMB 13966_ Ammonium transporter A7HT14

0.878 Denitrovibrio acetiphilus _strain DSM 12809 / NBRC 114555 / N2460_ Ammonium transporter D4H5F9

0.924

Desulfuromonas acetoxidans _strain DSM 684 / 11070_ Ammonium transporter Q1K433

0.968

0.67 Nematostella vectensis Ammonium transporter _Fragment_ A7SGD4

0.63 Branchiostoma floridae Ammonium transporter C3YB73

0.673

0.38

Nematostella vectensis Ammonium transporter _Fragment_ A7SQ16
Nematostella vectensis Ammonium transporter _Fragment_ A7RNC3 0.59 Nematostella vectensis Ammonium transporter A7S731

Nematostella vectensis Ammonium transporter A7S3L2

0.877

0.83

Lot ia gigantea Ammonium transporter _Fragment_ V3ZVE4

Lot ia gigantea Ammonium transporter V4CRP5 0.967

Branchiostoma floridae Ammonium transporter C3YB74

0.995

Aedes aegypti Ammonium transporter Q1L727

0.998 Bactrocera dorsalis Ammonium transporter A0A034WGQ2

0.935

Anopheles gambiae Ammonium transporter Q7Z1M1

0.87

Drosophila ananassae Ammonium transporter B3MX93

0.8

Drosophila kik awai Ammonium transporter A0A6P4JM51

0.88

Drosophila sechellia Ammonium transporter B4HDZ9

Drosophila melanogaster Ammonium transporter Q9VFA9

0.293

0.454

0.579
0.513

Desulfarculus baarsii _strain ATCC 33931 / DSM 2075 / LMG 7858 / VKM B-1802 / 2st14_ Ammonium transporter E1QIU3 0.555
Acetobacterium woodii _strain ATCC 29683 / DSM 1030 / JCM 2381 / KCTC 1655 / WB1_ Ammonium transporter H6LKX6

1

Desulfomicrobium baculatum _strain DSM 4028 / VKM B-1378 / X_ Ammonium transporter C7LW88

Desulfomicrobium norvegicum _strain DSM 1741 / NCIMB 8310_ Ammonium transporter A0A8G2F718 0.505 0.597

0.92

Daphnia pulex Ammonium transporter AmtB-like domain-containing protein E9H9L9
Daphnia pulex Ammonium transporter AmtB-like domain-containing protein E9FWC4

1

Cylindrospermum stagnale PCC 7417 Ammonium transporter K9WSX4

Anabaena cylindrica _strain ATCC 27899 / PCC 7122_ Ammonium transporter K9ZLY2

0.575

1

Moorella thermoacetica Y72 Ammonium transporter A0A0S6UHP2 Moorella thermoacetica Ammonium transporter A0A1J5NNB8 0.56
Chthonomonas calidirosea _strain DSM 23976 / ICMP 18418 / T49_ Ammonium transporter S0EUS1 0.624
Paludibacter propionicigenes _strain DSM 17365 / JCM 13257 / WB4_ Ammonium transporter E4T3F3

0.56

0.98

Pseudoalteromonas atlantica _strain T6c / ATCC BAA-1087_ Quaternary ammonium permease A0A9J9ETF7

Alteromonas mediter anea 615 Quaternary ammonium permease S5AG11

0.975

Methanosarcina barkeri 3 Ammonium transporter A0A0E3WZH5 0.76

Methanosarcina barkeri CM1 Ammonium transporter A0A0G3CAR5

Methanosarcina barkeri 227 Ammonium transporter A0A0E3R4G0 0.633

0.824 0.998

Faecalibacterium prausnitzi Ammonium transporter A0A3E2UCG9 0.823

Faecalibacterium prausnitzi Ammonium transporter A0A3E2T713 1

Faecalibacterium prausnitzi Ammonium transporter A0A3E2UHE06.46

Faecalibacterium prausnitzi Ammonium transporter A0A3E2VY88

Faecalibacterium prausnitzi Ammonium transporter A0A3E2TW06 0.893 0.99

Butyricicoc us pul icaecorum Adenylate cyclase A0A1Y4LM40. 97

Butyricicoc us pul icaecorum Adenylate cyclase A0A1Y4L5P0

Butyricicoc us pul icaecorum 1.2 Ammonium transporter R8VWI7 0.976

1

Roseburia hominis _strain DSM 16839 / JCM 17582 / NCIMB 14029 / A2-183_ Ammonium transporter G2T1Y3 Roseburia hominis Ammonium transporter A0A395VB65 0.937 Marvinbryantia formatexigens DSM 14469 Nitrogen regulatory protein P-I C6LBD3 0.86

Butyrivibrio fibrisolvens Adenylate cyclase A0A317G0I6 0.787 0.99

Pseudobutyrivibrio ruminis Adenylate cyclase A0A2G3E8V2

Pseudobutyrivibrio ruminis Adenylate cyclase A0A2G3DSU6 0.975
Anaerostipes hadrus ATCC 29173 _ JCM 17467 Nitrogen regulatory protein P-I L1PXI00.4 Anaerostipes hadrus Ammonia transporter A0A173R3X4 Anaerostipes hadrus Ammonia channel A0A6N2TIA4

0.877

Herpetosiphon aurantiacus _strain ATCC 23779 / DSM 785 / 1 4-95_ Ammonium transporter A9B874

0.954 0.87

Pirel ula staleyi _strain ATCC 27377 / DSM 6068 / ICPB 4128_ Ammonium transporter D2R647 Pelodictyon luteolum Ammonium transporter A0A165MFD1 0.977 0.94 Rhodopirel ula sal entina SM41 Ammonium transporter M5U2P1 Rhodopirel ula sp. SWK7 Ammonium transporter M5SYZ1 0.998 Rhodopirel ula europaea SH398 Ammonium transporter M5S9B0 4.958

Rhodopirel ula europaea 6C Ammonium transporter M2ALH90.997

Rhodopirel ula baltica SWK14 Ammonium transporter L7CG990.9

Rhodopirel ula baltica WH47 Ammonium transporter F2AUF0 4.58

Rhodopirel ula baltica SH28 Ammonium transporter K5DBA0
Rhodopirel ula baltica _strain DSM 10527 / NCIMB 13988 / SH1_ Ammonium transporter Q7UGU7

0.287

Coraliomargarita akajimensis _strain DSM 45221 / IAM 1541 / JCM 23193 / KCTC 12865 / 04OKA010-24_ Ammonium transporter D5ELM4

0.291

0.67

Desulfohalobium retbaense _strain ATCC 49708 / DSM 5692 / JCM 16813 / HR100_ Ammonium transporter C8WZE0 1 Bacteroides fragilis str. 2-F-2 #4 Ammonium transporter A0A015YIP3 1 Bacteroides fragilis Ammonium transporter A0A5M5PZE2 Bacteroides fragilis Ammonium transporter A0A642KQG8 0.27 0.694 0.53

Leptospira biflexa serovar Patoc _strain Patoc 1 / ATCC 23582 / Paris_ Ammonium transporter B0SRS8 0.9 Leptonema il ini DSM 21528 Ammonium transporter H2CDR5 Leptonema il ini Ammonium transporter A0A83 H2R7

0.938

1

Blastopirel ula marina Ammonium transporter A0A2S8GF54

Blastopirel ula marina DSM 3645 Ammonium transporter A3ZT81 0.92

Blastopirel ula marina Ammonium transporter A0A2S8F4W0 0.5

Blastopirel ula marina Ammonium transporter A0A2S8GEJ5 Blastopirel ula marina Ammonium transporter A0A2S8FTB5

0.303 0.95

Desulfatibacil um alkenivorans DSM 16219 Ammonium transporter A0A1M6ZMI3 Desulfatibacil um aliphaticivorans Ammonium transporter B8FL8 0.313

Desulfobotulus alkaliphilus Ammonium transporter A0A562RGZ4

0.317

0.43

Desulfuromonas acetoxidans _strain DSM 684 / 1 070_ Ammonium transporter Q1K3N8

0.56

Desulfobacter postgatei 2ac9 Ammonium transporter I5B3H0
Desulfobacula toluolica _strain DSM 7467 / Tol2_ Ammonium transporter K0NJH2

0.3

Melioribacter roseus _strain JCM 17 1 / P3M-2_ Ammonium transporter I6Z905 0.3 5

0.4

0.24

Ilyobacter polytropus _strain ATCC 512 0 / DSM 2926 / LMG 16218 / CuHBu1_ Ammonium transporter E3H8A6 Ilyobacter polytropus _strain ATCC 512 0 / DSM 2926 / LMG 16218 / CuHBu1_ Ammonium transporter E3H8C9

1

Geovibrio thiophilus Ammonium transporter A0A410K1J5
Denitrovibrio acetiphilus _strain DSM 12809 / NBRC 1 45 / N2460_ Ammonium transporter D4H2Y7

0.35

0.7 9

Archaeoglobus fulgidus Ammonium transporter A0A101DDH0

0.827

Kuenenia stut gartiensis Ammonium transporter Q1PX09

0.946

0.93

Methanohalobium evestigatum _strain ATCC BAA-1072 / DSM 3721 / NBRC 107634 / OCM 161 / Z-7303_ Ammonium transporter D7E934
Methanosalsum zhilinae _strain DSM 4017 / NBRC 107636 / OCM 62 / WeN5_ Ammonium transporter F7XMD8

0.973

Methanohalophilus mahi _strain ATCC 35705 / DSM 5219 / SLP_ Ammonium transporter D5E9 5 1 Methanosarcina barkeri 3 Ammonium transporter A0A0E3SHQ9 0.74 Methanosarcina barkeri 2 7 Ammonium transporter A0A0E3QYF3

Methanosarcina barkeri _strain Fusaro / DSM 804_ Ammonium transporter Q46 Y3

0.406

Caldithrix abys i DSM 13497 Ammonium transporter H1XV27

0.407

Proteiniphilum acetatigenes Ammonium transporter A0A101HKN9 0.409

0.8

Draconibacterium orientale Ammonium transporter X5DX97

0.89

Marinilabil a salmonicolor Ammonium transporter A0A2T0XPP2 Prolixibacter bel ari vorans Ammonium transporter A0A5M4AWD2 0.381

0.415

Desulfosporosinus orientis _strain ATCC 19365 / DSM 765 / NCIMB 8382 / VKM B-1628 / Singapore I_ Ammonium transporter G7WDB5

0.453 0.597

0.9

Butyrivibrio fibrisolvens Ammonium transporter A0A317FZW2

Butyrivibrio fibrisolvens DSM 3071 Ammonium transporter A0A1M5ZVP9

0.71

Eubacterium cal anderi Ammonium transporter E3GQ97
Acetobacterium wo di _strain ATCC 29683 / DSM 1030 / JCM 2381 / KCTC 165 / WB1_ Ammonium transporter H6LFQ2

0.82

Galbibacter orientalis DSM 19592 Ammonium transporter I3C0V3 1

Cel ulophaga lytica _strain ATCC 23178 / DSM 7489 / JCM 8516 / NBRC 14961 / NCIMB 1423 / VKM B-143 / Cy l20_ Ammonium transporter F0RHS0 0.72 Al omuricauda ruestringensis _strain DSM 13258 / CIP 107369 / LMG 19739 / B1_ Ammonium transporter G2PRJ6 Le uwenhoekiel a blandensis _strain CECT 71 8 / CCUG 51940 / KCTC 2 103 / MED217_ Ammonium transporter A3XNG6

0.3

0.691

0.1

Acidaminobacter hydrogenoformans DSM 2784 Ammonium transporter A0A1G5RZU9 0.06

Caldil nea aerophila _strain DSM 14535 / JCM 1 387 / NBRC 104270 / STL-6-O1_ Ammonium transporter I0I540 Guil ardia theta _strain CCMP2712_ Ammonium transporter L1JEX0

0.952

0.84

0.69

Ros el omorea marisflavi Ammonium transporter A0A165K0X5
Oceanobacil us iheyensis _strain DSM 14371 / CIP 107618 / JCM 1 309 / KCTC 3954 / HTE831_ Ammonium transporter Q8ERT9 1 Planococ us donghaensis MPA1U2 Ammonium transporter E7RJZ0 Planococ us donghaensis Ammonium transporter A0A1C7EL50 1 Lysinibacil us fusiformis Ammonium transporter A0A2I0UUY1 0.81 Lysinibacil us sphaericus Ammonium transporter A0A2S0JW39 Lysinibacil us sphaericus Ammonium transporter A0A2S5CTX8

0.434

0.64

Acetohalobium arabaticum _strain ATCC 49 24 / DSM 5 01 / Z-728 _ Ammonium transporter D9QS43

0.98

Halobacteroides halobius _strain ATCC 35273 / DSM 5150 / MD-1_ Ammonium transporter L0K9M1 Orenia marismortui Ammonium transporter A0A4R8H081 0.312 0.25

Dethiobacter alkaliphilus AHT 1 Ammonium transporter C0GKN4
Mahel a australiensis _strain DSM 15 67 / CIP 107919 / 50-1 BON_ Ammonium transporter F3ZYD6

0.351

0.438

0.845

Lot ia gigantea Ammonium transporter _Fragment_ V4AP67 0.9

Oikopleura dioica Ammonium transporter AmtB-like domain-containing protein E4XXT2

Oikopleura dioica Ammonium transporter AmtB-like domain-containing protein E4YA63

0.4

Dictyostelium purpureum Ammonium transporter F0ZPL3

0.17

Lot ia gigantea Ammonium transporter AmtB-like domain-containing protein V4A9D7

Dictyostelium discoideum Ammonium transporter 2 Q9BLG3

0.59

Petrotoga mobil s _strain DSM 10674 / SJ95_ Ammonium transporter A9BIZ7

0.58

1

Desulfosporosinus meridiei _strain ATCC BAA-275 / DSM 13257 / KCTC 12902 / NCIMB 13706 / S10_ Ammonium transporter J7IVQ1

Desulfosporosinus orientis _strain ATCC 19365 / DSM 765 / NCIMB 8382 / VKM B-1628 / Singapore I_ Ammonium transporter G7WJN0

0.52

Natronincola peptidivorans Ammonium transporter A0A1I0C9580.95

Formosa agariphila _strain DSM 15362 / KCTC 12365 / LMG 230 5 / KMM 3901 / M-2Alg 35-1_ Ammonium transporter T2KJ59 Kordia algicida OT-1 Ammonium transporter A9DQM8

0.904

0.874 0.903

0.96

Conexibacter woesei _strain DSM 14684 / CIP 108061 / JCM 1 494 / NBRC 10 937 / ID13157 _ Ammonium transporter D3F545

Conexibacter woesei _strain DSM 14684 / CIP 108061 / JCM 1 494 / NBRC 10 937 / ID13157 _ Ammonium transporter D3F0Z1

1

Rubrobacter xylanophilus Ammonium transporter A0A510HHK8
Rubrobacter xylanophilus _strain DSM 9 41 / NBRC 16129 / PRD-1_ Ammonium transporter Q1AUG3 0.8 1 1

Leptospiril um fer iphilum Ammonium transporter A0A094W9G3

Leptospiril um fer iphilum _strain ML-04_ Ammonium transporter J9ZB82

0.831

0.947

Bhargavaea cecembensis DSE10 Ammonium transporter M7NFH3

0.835

Caldalkalibacil us thermarum

_strain TA2.A1

_ Ammonium transporter F5L8M7

0.5

Halobacil us halophilus

_strain ATCC 35676 / DSM 2 6 / JCM 20832 / KCTC 3685 / LMG 17431 / NBRC 1024 8 / NCIMB 2 69

_ Ammonium transporter I0JI83

Cytobacil us kochi Ammonium transporter A0A248TDF2

0.8 1 0.625

Paenibacil us lactis 154 Ammonium transporter G4H971 0.98
Brevibacil us brevis

_strain 47 / JCM 6285 / NBRC 10 59

_ Ammonium transporter C0Z5S5

Brevibacil us brevis Ammonium transporter A0A2Z4MD56

0.987

Geobacil us kaustophilus

_strain HTA426

_ Ammonium transporter Q5KZ 9 0.908
Geobacil us kaustophilus Ammonium transporter A0A0D8BQX6 0.9 3
Anoxybacil us flavithermus AK1 Ammonium transporter M8E0R3 0.9 7

Anoxybacil us flavithermus Ammonium transporter A0A2G5RQK9 0.8 5

Anoxybacil us flavithermus NBRC 109594 Ammonium transporter R4FZE30.37

Anoxybacil us flavithermus

_strain DSM 21510 / WK1

_ Ammonium transporter B7GKP4
Anoxybacil us flavithermus Ammonium transporter A0A178TA08

0.974

0.961

Capsaspora owczarzaki

_strain ATCC 30864

_ Ammonium transporter A0A0D2VKX5

0.986

Bauldia litoralis Ammonium transporter A0A1G6ABK5

0.971

Methylococ

us capsulatus

_strain ATCC 3 0 9 / NCIMB 1

32 / Bath

_ Ammonium transporter family protein Q608B6 0.947
Laribacter hongkongensis Ammonium transporter family protein A0A248LNW8 0.918

0.781

Gal ionel a capsifer iformans

_strain ES-2

_ Ammonium transporter D9SI36

0.86

Aromatoleum aromaticum

_strain EbN1

_ Ammonium transporter Q5P1S8

0.94 0.45

Propionivibrio dicarboxylicus Ammonium transporter

_TC 1.A.1

_ A0A1G7UZX2

0.49

Leptothrix cholodni

_strain ATCC 51 68 / LMG 8142 / SP-6

_ Rh family protein/ammonium transporter B1Y842
Paramagnetospiril um magneticum

_strain ATCC 70 264 / AMB-1

_ Ammonia permease Q2W5V2

0.65

Methylophaga frap ieri

_strain ATCC BA -2434 / DSM 25690 / JAM7

_ Ammonium transporter I1YG38

0.4 7

0.34

Parvibaculum lavamentivorans

_strain DS-1 / DSM 13023 / NCIMB 1396

_ Rh family protein/ammonium transporter A7HSN9
Mariprofundus fer ooxydans PV-1

Putative ammonium transporter Q0F2M5

0.92

Marichromatium purpuratum 984 Ammonium transporter W0DZX1
Al ochromatium vinosum

_strain ATCC 1789

/ DSM 180 / NBRC 103801 / NCIMB 104

1 / D

_ Ammonium transporter D3RN2

0.719 0.15

Herbaspiril um rubrisubalbicans Os34 Ammonium transporter A0A6M3ZKU5
Methyloversatilis universalis

_strain ATCC BA -1314 / DSM 25237 / JCM 13912 / CCUG 52030 / FAM5

_ Ammonium transporter F5R974

0.708 0.797

Halothiobacil us neapolitanus

_strain ATCC 23641 / c2

_ Ammonium transporter D0KXQ0 1

Acidithiobacil us fer ivorans Ammonium transporter A0A257TE48 0.925

Acidithiobacil us fer ivorans

Putative ammonium transporter A0A060UN760.95

Acidithiobacil us fer ivorans SS3 Ammonium transporter G0JRQ3

Acidithiobacil us fer ivorans Ammonium transporter A0A1B9BUR6

0.789

Hahel a chejuensis

_strain KCTC 2396

_ Ammonia permease Q2SIR0

0.788

Reinekea blandensis MED297 Ammonia permease A4BEP1

0.775

0.89

Enterovibrio norvegicus FF-454 Ammonium transporter A0A1E5C7W4 Enterovibrio norvegicus Ammonium transporter A0A2N7L7C8 0.733 0.4

Moritel a marina ATCC 15381

Ammonium transporter A0A5J6WPP9

Agarivorans albus MKT 106 Ammonium transporter

R9PMC6

0.612

0.68

Pseudoalteromonas atlantica

_strain T6c / ATCC BAA-1087

_ Ammonium transporter A0A9J9ES23

0.96

Neptuni bacter caesariensis

Putative ammonium permease A0A7U8C2Y6
Neptuni bacter caesariensis Ammonium transporter A0A2G6JLT9

0.413

0.99

Photobacterium profundum

_strain SS9

_ Ammonium permease

Q6LKC8

Photobacterium profundum 3TCK

Putative ammonium permease

Q1Z695

0.51

0.21

Marinomonas mediter anea

_strain ATCC 700492 / JCM 21426 / NBRC 103028 / MMB-1

_ Ammonium transporter

F2JZ83

Psychromonas ingrahami

_strain DSM 17664 / CCUG 51855 / 37

_ Ammonium transporter A1SVA8

0.625

Acinetobacter haemolyticus ATCC 19194 Ammonium transporter

D4XSW0

0.5

Orenia marismortui Ammonium transporter A0A4R8GSH7

Desulfurobacterium thermolithotrophum

_strain DSM 11699 / BSA

_ Ammonium transporter

F0S3Z6

0.965

0.775

0.65

Emiliania huxleyi Ammonium transporter

_Fragment

_ R1CIA4

Nitrososphaera viennensis Ammonium transporter A0A977NL98

0.785

Naegleria gruberi Ammonium transporter

D2V1J4

0.87

Dictyostelium

purpureum Ammonium transporter

F0ZWG6

Dictyostelium discoideum Ammonium transporter 1

Q9BLG4

0.427

0.433

0.66

Oscil ochloris trichoides DG-6 Ammonium transporter

E1IF27

Herpetosiphon aurantiacus

_strain ATCC 23779 / DSM 785 / 114-95

_ Ammonium transporter A9AUR8

0.442

0.452

1 0.647

Pelotomaculum thermopropionicum

_strain DSM 13744 / JCM 10971 / SI

_ Ammonium transporter A5D4I5

0.935

Propionispira arboris Ammonium transporter A0A1H7CU41

0.72

Mitsuokel a multacida Ammonium transporter A0A414NWT8

Mitsuokel a

multacida DSM 20544 Ammonium transporter

C9KKP2

0.644

Acetonema longum DSM 6540 Ammonium transporter

F7NPQ3

0.705

Mahel a australiensis

_strain DSM 15567 / CIP 107919 / 50-1

BON

_ Ammonium transporter

F3ZVV4

0.802

Ammonifex degensi

_strain DSM 10501 / KC4

_ Ammonium transporter

C9R9J0

0.95
0.86

Moorel a thermoacetica

Ammonium transporter A0A1J5NTP7
Moorel a thermoacetica

_strain ATCC 39073 / JCM 9320

_ Ammonium transporter

Q2RM12 0.765

Moorel a thermoacetica

Ammonium transporter A0A1J5JH24

0.56

Moorel a thermoacetica Y72

Ammonium transporter A0A0S6UEK9

Moorel a thermoacetica

Ammonium transporter A0A853W4S3

0.468

0.804

Desulfobacca

acetoxidans

_strain ATCC 700848 / DSM 11109 / ASRB2

_ Ammonium transporter

F2NGJ8
1

Dehalococcoides

mccartyi

_strain VS

_ Ammonium transporter

D2BIB2

0.76

0.93

Dehalococcoides

mccartyi

Ammonium transporter A0A2J1DY21
Dehalococcoides

mccartyi

Ammonium transporter A0A328EMN2

0.58

Dehalococcoides

mccartyi

Ammonium transporter A0A0V8M0W6

Dehalococcoides

mccartyi

_strain ATCC BAA-2266 / KCTC 15142 / 195

_ Ammonium transporter

Q3Z7F9

0.547

Desulfobulbus

propionicus

_strain ATCC 33891

/ DSM 2032 / VKM B-1956 / 1pr3

_ Ammonium transporter

A0A7U4DN04

0.546

0.63

0.7

Desulfomonile tiedjei

_strain ATCC 49306 / DSM 6799 / DCB-1

_ Ammonium transporter

I4C9L2

Desulfomonile tiedjei

_strain ATCC 49306 / DSM 6799 / DCB-1

_ Ammonium transporter

I4C9L0

0.915

Desulfobotulus

alkaliphilus

Ammonium transporter

A0A562S8V2 1

Desulfatibacil

um

alkenivorans DSM

16219

Ammonium transporter

A0A1M6BPM0

Desulfatibacil

um

aliphaticivorans

Ammonium transporter

B8FDY8

0.609

0.84

Pseudodesulfovibrio

mercuri

Ammonium transporter

F0JGZ1

1

Desulfovibrio desulfuricans

Ammonium transporter

A0A4P7UHQ5 1

Desulfovibrio desulfuricans

_strain ATCC 27774 / DSM 6949 / MB

_ Ammonium transporter

B8IZX4

Desulfovibrio desulfuricans

Ammonium transporter

A0AA94HTL8

0.604

Desulfomicrobium

baculatum

_strain

DSM 4028 / VKM

B-1378 / X

_ Ammonium transporter

C7LW87

0.537

Desulfomicrobium

norvegicum

_strain

DSM

1741

/ NCIMB 8310

_ Ammonium transporter

A0A8G2F3G4

0.38

Desulfohalobium

retbaense

_strain ATCC 49708 / DSM

5692 / JCM

16813 / HR100

_ Ammonium transporter

C8X554

0.17

Desulfomonile tiedjei _strain ATCC 49306 / DSM

6799 / DCB-1

_ Ammonium transporter

I4CCD1

0.13

Bilophila

wadsworthia

_strain

3

_1

_6

_ Ammonium transporter

E5Y789

Pseudodesulfovibrio

piezophilus

_strain

DSM

21447 / JCM

15486 / C1TLV30

_ Ammonium

transporter

M1WRK6

0.816

Chthonomonas

calidirosea

_strain

DSM

23976 / ICMP

18418 / T49

_ Ammonium

transporter

S0ET45

0.853

Desulfarculus

baarsi _strain ATCC 33931

/ DSM

2075 / LMG

7858 / VKM

B-1802 / 2st14

_ Ammonium

transporter

E1QFA8

0.895

Ktedonobacter

racemifer DSM

44963

Ammonium

transporter

D6TCH6

0.93

0.496

Archaeoglobus fulgidus

DSM

8774

Ammonium

transporter

A0A075WEB4

0.468

1 Ignicoccus hospitalis _strain KIN4/I / DSM 18386 / JCM 14125_ Ammonium transporter A8AC17 Pyrolobus fumarii _strain DSM 11204 / 1A_ Ammonium transporter G0EFF3

0.745

Ferroglobus

placidus

_strain

DSM

10642 / AEDII12DO

_

Ammonium

transporter

D3S2L0

1

Archaeoglobus fulgidus

DSM

8774

Ammonium

transporter

A0A075WBP9

Archaeoglobus fulgidus

Ammonium

transporter

_Fragment

_

A0A101DCS3

0.795

Methanosphaerula

palustris

_strain ATCC

BAA-1556 /

DSM

19958 /

E1-9c

_

Ammonium

transporter

B8GGC1

0.769

0.503

Methanoregula

boonei _strain

DSM

21154 / JCM

14090 / 6A8

_

Ammonium

transporter

A7I8Z2

0.68

Methanocorpusculum labreanum

_strain ATCC

43576 /

DSM

4855 / Z

_

Ammonium

transporter

A2SPS8

0.99

Methanospirillum

hungatei Ammonium

transporter

A0A8F5ZDT8

Methanospirillum

hungatei JF-1

_strain ATCC

27890

/

DSM

864

/

NBRC

100397

/ JF-1

_

Ammonium

transporter

Q2FS34

0.894

Methanosphaerula

palustris

_strain ATCC

BAA-1556

/

DSM

19958

/

E1-9c

_

Ammonium

transporter

B8GGC3

0.987

Methanoregula

boonei _strain

DSM

21154

/ JCM

14090

/

6A8

_

Ammonium

transporter

A7I8Z4

0.924

Methanofollis liminatans

DSM

4140

Ammonium

transporter

J1L5M8

0.858

Methanofollis liminatans

DSM

4140

Ammonium

transporter

J1ASR8

0.83

Methanomethylovorans

hollandica

_strain

DSM

15978

/

NBRC

107637

/

DMS1

_

Ammonium

transporter

L0KW14

0.93

Methanocorpusculum labreanum

_strain ATCC

43576

/

DSM

4855

/

Z

_

Ammonium

transporter

A2SS52

1

Methanospirillum

hungatei JF-1

_strain ATCC

27890

/

DSM

864

/

NBRC

100397

/

JF-1

_

Ammonium

transporter

Q2FQT3

Methanospirillum

hungatei Ammonium

transporter

A0A8F5VN39

0.882

Chondromyces

apiculatus

DSM

436

Ammonium

transporter

A0A017SYR8 0.8

81 0.928 0.733

0.99

Nitrososphaera

viennensis

Ammonium

transporter

A0A977NM15

Nitrososphaera

viennensis

EN76

Ammonium

transporter

A0A060HSP0

1

Cenarchaeum

symbiosum

_strain A

_

Ammonia

permease

A0RXK8

Cenarchaeum

symbiosum

_strain A

_

Ammonium

transporter

A0RUZ2

0.974

0.574

1

Cryptococcus

neoformans

var. neoformans

serotype

D

_strain

JEC21

/ ATCC

MYA-565

_

Ammonium

transporter

Q5KPM5 0.92

Cryptococcus

neoformans

var. grubii serotype A

_strain

H99

/ ATCC

208821

/

CBS

10515

/

FGSC

9487

_

Ammonium

transporter

J9VID9

Cryptococcus

neoformans

var. grubii c45

Ammonium

transporter

A0A226A1

N0

1

Allomyces

macrogynus

_strain ATCC

38327

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Ammonium

transporter

A0A0L0TDW1

1

Allomyces

macrogynus

_strain

ATCC

38327

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Ammonium

transporter

A0A0L0S493

Allomyces

macrogynus

_strain

ATCC

38327

_

Ammonium

transporter

A0A0L0STS6

0.91

9

0.625

Coemansia

reversa

_strain

ATCC

12441

/

NRRL

1564

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Ammonium

transporter

A0A2G5BGA9

0.638

0.455

Debaryomyces

hansenii _strain

ATCC

36239

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CBS

767

/

BCRC

21394

/

JCM

1990

/

NBRC

0083

/ IGC

2968

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DEHA2F24442p

Q6BK68

0.78

Coemansia

reversa

_strain

ATCC

12441

/

NRRL

1

564

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Ammonium

transporter

A0A2G5BI87

Batrachochytrium

dendrobatidis

_strain

JAM81

/

FGSC

10211

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Ammonium

transporter

F4NRY2

0.779

Lodderomyces

elongisporus

_strain

ATCC

11

503

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CBS

2605

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JCM

1

781

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NBRC

1676

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NRRL

YB-4239

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Ammonium

transporter

A5E3X2

0.661

Candida

albicans

_strain

SC5314

/ ATCC

MYA-2876

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Ammonium

transporter

Q59UP8

0.633

Debaryomyces

hansenii _strain

ATCC

36239

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CBS

767

/

BCRC

21

394

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JCM

1

990

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NBRC

0083

/ IGC

2968

_

Ammonium

transporter

Q6BT62

0.732
0.14

Neurospora

crassa

_strain

ATCC

24698

/

74-OR23-1A /

CBS

708.71

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DSM

1

257

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FGSC

987

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Ammonium

transporter

Q7S5K0

Saccharomyces

cerevisiae

_strain

ATCC

204508

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S288c

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Ammonium

transporter

MEP2

P41

948

0.96

Naumovozyma

castellii _strain

ATCC

76901

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BCRC

22586

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CBS

4309

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NBRC

1

992

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NRRL

Y-1

2630

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Ammonium

transporter G0VI62

0.74

Eremothecium

cymbalariae

_strain

CBS

270.75

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DBVPG

721

5

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KCTC

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71

66

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NRRL

Y-1

7582

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Ammonium

transporter G8JUJ2

0.98

Kluyveromyces lactis

_strain

ATCC

8585

/

CBS

2359

/

DSM

70799

/

NBRC

1

267

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NRRL

Y-11

40

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Ammonium

transporter Q6CRN0

Kluyveromyces lactis

_strain

ATCC

8585

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NRRL

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Ammonium

transporter Q6CRN1

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Neurospora

crassa

Ammonium

transporter Q8X0D3

0.733

0.995

Emericella

nidulans

_strain

FGSC

A4

/ ATCC

381

63

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2.46

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NRRL

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Ammonium

transporter G5EB1

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0.98

Aspergillus

clavatus

_strain

ATCC

1

007

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CBS

51

3.65

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DSM

81

6

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NCTC

3887

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NRRL

1

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QM

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Ammonium

transporter A1

CQ21

0.985

Neosartorya

fischeri _strain

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Ammoniu

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D2Y50.62

Aspergillus

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MYA-4609

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Am

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m transporter Q4WT1

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Aspergillus

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Am

moniu

m transporter A0A8H4GS02

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Laccaria

bicolor _strain

S238N-H82

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ATCC

MYA-4686

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Am

moniu

m transporter B0DFJ4

Hypholoma

sublateritiu

m

_strain

FD-334

SS-4

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Am

moniu

m transporter A0A0D2NYP3 0.997
Cryptococcus

neoformans

var. neoformans

serotype

D

_strain

JEC21 /

ATCC

MYA-565

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Am

moniu

m transporter Q5KAF60.995

Cryptococcus

neoformans

var. grubii serotype

A

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ATCC

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CBS 10515

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Am

moniu

m transporter J9VXX90.79
Cryptococcus

neoformans

var. grubii c45

Am

moniu

m transporter A0A225XW14
Cryptococcus

neoformans

var. grubii Tu259-1 Am

moniu

m transporter A0A854Q6X9

0.793

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0.61 Beauveria bassiana Ammonium transporter A0A2N6P374

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bassiana

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Am

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bicolor _strain

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orter E4

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5Z1 Marinithermus hydrothermalis _strain DSM 14884 / JCM 11576 / T1_ Ammonium transporter F2NNQ0

0.398 1 Anaeromyxobacter dehalogenans _strain 2CP-1 / ATCC BAA-258_ Ammonium transporter B8JBH5 Anaeromyxobacter dehalogenans _strain 2CP-C_ Ammonium transporter Q2IHY5 0.507 Gloeobacter violaceus _strain ATCC 29082 / PCC 7421_ Ammonium transporter Q7NCB8 0.65

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T-2

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M 11422 / N

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Aca

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castella

nii _

strain

ATCC
30010 / N

eff_

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niu

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bfa

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L8G

FX3

Aca

ntha

m

oe

ba
castella

nii _

strain

ATCC
30010 / N

eff_

Am

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m tra

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porter L8G

GA0

0.779 1 Alicyclobacillus acidocaldarius subsp. acidocaldarius _strain ATCC 27009 / DSM 446 / BCRC 14685 / JCM 5260 / KCTC 1825 / NBRC 15652 / NCIMB 11725 / NRRL B-14509 / 104-IA_ Ammonium transporter C8WSI9 Alicyclobacillus acidocaldarius _strain Tc-4-1_ Ammonium transporter F8IJE0 0.91 Pseudoramibacter alactolyticus ATCC 23263 Ammonium transporter E6ME92 0.997 Denitrobacterium detoxificans Ammonium transporter A0A172RYW3 0.927 Enorma massiliensis Ammonium transporter A0A1Y3U5V2 0.9 0.884 9 Adlercreutzia equolifaciens subsp. celatus Ammonium transporter A0A369NYJ2 Adlercreutzia equolifaciens subsp. celatus DSM 18785 Ammonium transporter A0A3N0AQL8 1 Eggerthella lenta _strain ATCC 25559 / DSM 2243 / CCUG 17323 / JCM 9979 / KCTC 3265 / NCTC 11813 / VPI 0255 / 1899 B_ Ammonium transporter C8WLL8 0.825 Gordonibacter pamelaeae 7-10-1-b Ammonium transporter D6E7B3 1 Eggerthella lenta Ammonium transporter A0A369MHK2 0.615 Eggerthella lenta Ammonium transporter A0A369N7Y3 0.62 Eggerthella lenta Ammonium transporter A0A5C5BWI6 Eggerthella lenta Ammonium transporter A0A369MRW6

0.901

0. 0.407 96

Gordonia
bronchialis

_
strain
ATCC

25592

/
DSM
43247

/

BCRC 13721 /

JCM

3198

/
KCTC

3076

/

N
BRC 16047

/

NCTC 10667_

Am
moniu

m transporter D0L8H3

0.99

Pseudoclavibacter sp. RFBB5

Am
moniu

m transporter A0A2S5WU

P60.72

Pseudoclavibacter sp. RFBG4

Am
moniu

m transporter A0A2S5WGI8

Pseudoclavibacter sp. AY1F1 Am

moniu

m transporter A0A2S5V2F6
0.445

Kitasatospora

setae

_

strain
ATCC

33774

/
DSM
43861

/
JCM

3304

/
KCC
A-0304

/
NBRC

1

421

6

/

KM-6054_

Am
moniu

m transporter E4N

1

T5

0.464
0.09

Fim
briimonas

ginsengisoli Gsoil 348

Am
moniu

m transporter A0A068NPW2

Acidotherm

us

cellulolyticus

_

strain
ATCC
43068

/
DSM

8971

/

1

1
B_

Am
moniu

m transporter A0LV78

0.627

1

Micavibrio
aeruginosavorus

Am
moniu

m transporter A0A2W4ZXC0

Micavibrio
aeruginosavorus

EPB
Ammoniu

m transporter M4VEJ9

0.65

Deinococcus
geothermalis

_

strain

DSM

1

1
300
/ AG-3a_

Ammoniu

m

transporter Q1IWN6

Cyanobiu

m
gracile

_

strain
ATCC

271

47

/
PCC
6307_
Ammoniu

m

transporter K9P5D1

0

0.896 .92

Nocardia
brasiliensis

_

strain
ATCC
700358

/

HUJEG-1_
Ammonium
transporter K0ER80

Nocardia
brasiliensis
Ammonium

transporter A0A6G9XJV9

0.879

0.635

Kribbella

flavida

_

strain
DSM

1
7836

/
JCM

1
0339

/
NBRC

1

4399_
Ammonium
transporter D2Q0J8

0.91

Marinactinospora
thermotolerans

DSM

451

54

Ammonium
transporter A0A1

T4SFY6

Nocardiopsis

alba

_

strain
ATCC
BAA-21

65

/
BE74_
Ammonium
transporter J7L3W8

0.856

0.59

Aeromicrobium

marinum

DSM

1
5272
Ammonium
transporter
E2SD33

0.51

3

0.83

Kutzneria

albida
DSM
43870
Ammonium
transporter
W5WIX6

Amycolatopsis
mediterranei _

strain
U-32_
Ammonium
transporter
A0A0H3D0Q9

0.9

Catenulispora
acidiphila

_

strain
DSM
44928

/
JCM
14897

/
NBRC

1
021

08

/
NRRL
B-24433

/ ID1
39908_
Ammonium
transporter

C7Q1

E4

Catenulispora
acidiphila

_

strain
DSM
44928

/
JCM
14897

/
NBRC

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021

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NRRL
B-24433

/ ID1
39908_
Ammonium
transporter
C7QGX3

0.83

Actinomadura

madurae
Ammonium
transporter
A0A1I4W3A0

0.827

1

Kineosphaera limosa

NBRC
100340
Ammonium
transporter
K6W5X0

Mobilicoccus
pelagius
NBRC
104925
Ammonium
transporter
H5UND3

0.804

0.63

0.601

0.54

Nakamurella
multipartita

_

strain
ATCC
700099

/
DSM
44233

/
CIP
104796

/
JCM
9543

/
NBRC
105858

/

Y-104_
Ammonium
transporter
C8XB51

Dietzia
cinnamea
Ammonium
transporter
A0A4R3ZNW3

0.722

0.99

Nocardia
brasiliensis

_

strain
ATCC
700358

/

HUJEG-1_
Ammonium
transporter
K0F9X7

Nocardia
brasiliensis
Ammonium
transporter
A0A6G9Y3Q0

0.71

0.9

Actinopolyspora

mortivallis
Ammonium
transporter
A0A2T0GTU1

Saccharopolyspora

erythraea

_

strain
ATCC
11635

/
DSM
40517

/
JCM
4748

/

NBRC
13426

/

NCIMB

8594

/
NRRL
2338_
Ammonium
transporter
A4FMG0

0.655

Pseudonocardia
dioxanivorans

_

strain
ATCC
55486

/
DSM
44775

/
JCM
13855

/

CB1190_
Ammonium
transporter
F4D1J0

0.97

Actinosynnema

mirum

_

strain
ATCC
29888

/
DSM
43827

/
JCM
3225

/

NBRC
14064

/

NCIMB
13271

/
NRRL
B-12336
/ IMRU

3971

/

101_
Ammonium
transporter
C6WEW7

Saccharothrix
espanaensis

_

strain
ATCC
51144

/
DSM
44229

/
JCM
9112

/

NBRC
15066

/
NRRL
15764_
Ammonium
transporter
K0K7J4

0.778

Prauserella

rugosa
Ammonium
transporter
A0A660CH85

0.803

0.83

1

Kocuria
sp. ChxB
Ammonium
transporter
A0A9Y2AG30

Kocuria
rhizophila

_

strain
ATCC
9341

/
DSM

348

/

NBRC
103217

/

DC2201_
Ammonium
transporter
B2GFN1
0.99

Micrococcus luteus
Ammonium
transporter
A0A653ISC0

0.987

Micrococcus luteus
Ammonium
transporter
A0A378NLG8 0.89

Micrococcus luteus

SK58

Ammonium
transporter
D3LL06

0.45

Micrococcus
aloeverae
Ammonium
transporter
A0AA46ESZ3

Micrococcus
yunnanensis
Ammonium
transporter
A0A4Y8PPH8

0.739

Nakamurella
multipartita

_

strain
ATCC
700099

/
DSM
44233

/
CIP
104796
/ JCM
9543

/

NBRC
105858
/ Y-104_
Ammonium
transporter
C8XBX8

0.731
0.493

Actinotalea
ferrariae
CF5-4
Ammonium
transporter
A0A021VMC6

0.715

Bifidobacterium

animalis
subsp. lactis
Ammonium
transporter
A0A315S845

1

Alloscardovia
omnicolens
Ammonium
transporter
A0A2I1M481

Alloscardovia
omnicolens

F0580
Ammonium
transporter
U1SIR9

0.698

Cellulomonas
flavigena _

strain
ATCC

482

/
DSM
20109 /
BCRC

11376 / JCM
18109 /
NBRC
3775 /
NCIMB
8073 /
NRS
134_
Ammonium
transporter
D5UD26

0.69

Beutenbergia
cavernae _

strain
ATCC
BAA-8 /

DSM

12333 / CCUG

43141
/ JCM
11478 /
NBRC
16432

/

NCIMB
13614 /
HKI 0122_
Ammonium
transporter
C5BWX7

0.71

0.94

Dermacoccus
nishinomiyaensis
Ammonium
transporter
A0A075JIN0 Intrasporangium calvum _strain ATCC 23552 / DSM 43043 / JCM 3097 / NBRC 12989 / 7 KIP_ Ammonium transporter E6SF19

0.787 Isoptericola variabilis _strain 225_ Ammonium transporter F6FRU9 0.807

Jonesia

denitrificans _
strain ATCC
14870 /

DSM
20603 /
BCRC
15368 / CIP
55.134 / JCM

11481

/

NBRC
15587 /
NCTC
10816 /

Prevot 55134_
Ammonium
transporter
C7R314

0.83

Paraoerskovia

marina

Ammonium
transporter
A0A1H1U2Q6

1

Cellulosimicrobium

cellulans
Ammonium
transporter
A0A4Y4DZN4

Cellulosimicrobium

cellulans
Ammonium
transporter
A0A1Y0HX98

0.759

0.523
0.857

Microlunatus
phosphovorus _
strain ATCC
700054 / DSM
10555 / JCM
9379 / NBRC
101784 / NCIMB

13414 / VKM Ac-1990 / NM-1_

Ammonium
transporter
F5XN43

0.99

Geodermatophilus

obscurus
Ammonium
transporter
A0A1I5GN22

0.91

Geodermatophilus

obscurus
Ammonium
transporter
A0A1M7UEG2

Geodermatophilus
obscurus _

strain ATCC 25078 / DSM 43160 / JCM
3152 / KCC A-0152 / KCTC 9177 / NBRC

13315 / NRRL
B-3577 / G-20_
Ammonium
transporter
D2SE15

0.665
0.915

Kineococcus
radiotolerans _
strain ATCC
BAA-149 / DSM
14245 / SRS30216_

Ammonium
transporter
A6W7U2

0.98

Blastococcus
saxobsidens
Ammonium
transporter
A0A4Q7Y9I7

Blastococcus
saxobsidens _

strain
DD2_
Ammonium
transporter
H6RWK9

0.571

0.94

Gordonia
bronchialis _

strain ATCC 25592 / DSM 43247 / BCRC

13721
/ JCM

3198 / KCTC 3076 / NBRC

16047 / NCTC
10667_
Ammonium
transporter
D0LAL7

Dietzia
cinnamea
Ammonium
transporter
A0A4R3ZZ35
0.685

Leifsonia

xyli subsp. cynodontis
DSM 46306
Ammonium
transporter
U3P8G1

0.68

0.96

Brevibacterium

sp. CS2
Ammonium
transporter
A0A4P8HC89

1

Brevibacterium

linens

Ammonium
transporter
A0A142NNN1

Brevibacterium
linens ATCC 9172
Ammonium
transporter
A0A2H1HV86

0.665

Microbacterium paraoxydans

Ammonium
transporter
A0A1H1MB73

0.593

0.98

Leucobacter chromi resistens

Ammonium
transporter
A0A1H1BPD6

Leucobacter chromi resistens

Ammonium
transporter
A0A147EQL9 0.06

Salinibacterium
amurskyense
Ammonium

transporter family
A0A2M9D989

Salinibacterium
amurskyense
Ammonium
transporter
A0A2M9D7S8

0.75

0.815

Actinoplanes
missouriensis _

strain ATCC 14538 / DSM 43046 / CBS 188.64 / JCM 3121

/ NBRC 102363 / NCIMB

12654 / NRRL B-3342 / UNCC 431_

Ammonium
transporter
I0HGY4

1

Salinispora
arenicola _

strain

CNS-205_
Ammonium
transporter
A8M670

Salinispora
arenicola
Ammonium
transporter
A0A542XGT4
0.818

Rhodoluna
lacicola

Ammonium
transporter
A0A060JMP6

0.804

0.867

Agrococcus
sediminis
Ammonium
transporter
A0A5M8QF12 0.995

Aeromicrobium
marinum
DSM
15272

Ammonium
transporter
E2SD44

1

Hylemonel a gracilis str. Niagara

R

Ammonium
transporter
A0A016XDZ5

Hylemonel a gracilis
Ammonium
transporter
A0A4P6UIJ8

0.76

0.853

Microbacterium

sp. T32

Ammonium
transporter
A0A161SKF4

0.99

0.76

Pseudoclavibacter sp. RFBB5

Ammonium
transporter
A0A2S5WV24

Pseudoclavibacter sp. RFBA6

Ammonium
transporter
A0A2S5XAB8

0.99

Pseudoclavibacter sp. RFBI5
Ammonium transporter
A0A8E2UDJ3

Pseudoclavibacter sp. RFBG4
Ammonium transporter
A0A2S5WZU0

0.661
0.83

1

0.9

Pseudoclavibacter sp. RFBA6
Ammonium transporter
A0A2S5X482

Pseudoclavibacter sp. RFBB5
Ammonium transporter
A0A2S5WMT7

0.99

Pseudoclavibacter sp. RFBI5 Ammonium transporter

A0A8E2R1Y5

Pseudoclavibacter sp. RFBG4 Ammonium transporter

A0A2S5WX34

0.76

Rathayibacter festucae

DSM 15932 Ammonium transporter

A0A3T0SZ22

0.78

0.63

Rathayibacter sp. AY1B1
Ammonium transporter
A0A2S5YID7

Rathayibacter sp. AY1F3 Ammonium transporter

A0A2S5XMS2

0.79

Rathayibacter rathayi Ammonium transporter

A0A2S5Y0D3 0.835

Rathayibacter tritici Ammonium transporter

A0A160KTE9

0.71

Rathayibacter caricis DSM 15933 Ammonium transporter

A0A2T4UWF1

Curtobacterium flaccumfaciens
Ammonium transporter
A0A4R6DFU8

0.989

1

Clavibacter michiganensis
Ammonium transporter
A0A251YFS0

Clavibacter michiganensis subsp. tessel arius
Ammonium transporter
A0A225CMT8

0.831

Clavibacter michiganensis
Ammonium transporter
A0A251XU71

0.848

Clavibacter michiganensis
Ammonium transporter
A0A251YU48

0.806

Clavibacter lycopersici Ammonium transporter

A0A399T909

0.69

Clavibacter michiganensis
Ammonium transporter
A0A251Y3N4

0.717

Clavibacter sepedonicus
Ammonium transporter

B0RC35

0.895

Clavibacter michiganensis
Ammonium transporter
A0A399NT81

0.98

Clavibacter michiganensis subsp. michiganensis _

strain NCPPB 382_
Ammonium transporter

A5CRL2

Clavibacter michiganensis subsp. michiganensis

Ammonium transporter
A0A251XJT0

0.791

0.983

1

Desulfuromonas acetoxidans _
strain DSM 684 / 11070_
Ammonium transporter

Q1JXS5

1

Desulfatibacil um aliphaticivorans
Ammonium transporter

B8FGW0

Desulfatibacil um alkenivorans DSM 16219 Ammonium transporter A0A1M6JKY3

0.828

0.49

Prosthecochloris aestuari _
strain DSM 271 / SK 413_
Ammonium transporter

B4S394

Chlorobaculum parvum _
strain DSM 263 / NCIMB 8327_
Ammonium transporter

B3QRJ4
0.795

Chlorobium chlorochromati _

strain CaD3_

Ammonium transporter

Q3AP28

0.96

Pelodictyon luteolum

Ammonium transporter A0A165MAC2

Chlorobium luteolum _

strain DSM 273 / BCRC 81028 / 2530_
Ammonium transporter

Q3B6Q2

0.79

0.848

Metal osphaera sedula _

strain ATCC 51363 / DSM 5348 / JCM 9185 / NBRC 15 09 / TH2_

Ammonium transporter A4YGZ2

0.856

0.98

Leifsonia xyli subsp. xyli Ammonium transporter A0A1E2SKM1

Leifsonia xyli subsp. cynodontis DSM 46306 Ammonium transporter AmtB-like domain-containing protein

U3PCH5

0.916

0.86

Caldivirga maquilingensis _
strain ATCC 70 84

/ DSM 13496 / JCM 10307 / IC-167_
Ammonium transporter A8MCL4
0.985

Fer oplasma acidarmanus Fer1 Ammonium transporter

S0APH6

0.9

Acidianus hospitalis Ammonium transporter A0A2T9X1V4

Acidianus hospitalis _

strain W1_

Ammonium transporter

F4B548

0.98

Acidimicrobium fer ooxidans _

strain DSM 103

1 / JCM 15462 / NBRC 1038 2 / ICP_
Ammonium transporter

C7M2A9

0.96

0.9

Acidithiobacil us fer ivorans Ammonium transporter A0A257T8W6

0.93

Acidithiobacil us fer ivorans Ammonium transporter A0A060UWC8 Acidithiobacil us fer ivorans SS3 Ammonium transporter G0JRG6

0.963

0.86

Oryza sativa subsp. japonica Ammonium transporter 2 member 1

Q84KJ7

0.796

0.753

Ricinus communis Ammonium transporter

B9SXC7

0.575

Arabidopsis thaliana Ammonium transporter 2 Q9M6N7
0.4

Pyrus betulifolia Ammonium transporter

X5CNT2

Ricinus communis Ammonium transporter

B9SWW9

0.783

Oryza sativa subsp. japonica Ammonium transporter 3 member 1 Q84KJ6

0.985

Ricinus communis Ammonium transporter

B9SA70

0.98

Populus trichocarpa Ammonium transporter A0A2K1RA7 Populus trichocarpa Ammonium transporter A0A2K1RA81

0.909

Ricinus communis Ammonium transporter

B9RP82

0.965

Acidiphilium multivorum _

strain DSM 1 245 / JCM 8 67 / NBRC 10 8 3 / AIU 301_

Ammonium transporter

F0J7D3

0.919

0.9

Leptospiril um fer iphilum YSK Ammonium transporter A0A059X R1 Leptospiril um fer iphilum _strain ML-04_ Ammonium transporter J9Z839

0.908

Acidithiobacil us fer ivorans Ammonium transporter A0A257SYX3

0.948

Acidithiobacil us fer ivorans Ammonium transporter A0A257T7G4

0.973

Acidithiobacil us fer ivorans Ammonium transporter A0A7T4WEP4

0.6 5

Acidithiobacil us fer ivorans S 3 Ammonium transporter G0JRG9

1

Acidithiobacil us fer ivorans Ammonium transporter A0A060URC3 Acidithiobacil us fer ivorans Ammonium transporter A0A1B9BUU6

0.807

1

Maricaulis maris _
strain MCS10_

Ammonium transporter Q0ARJ4

Maricaulis maris Ammonium transporter A0A495DND0

0.804

0.713

0.495

0.75

Edaphobacter ag

regans Ammonium transporter A0A3R9Q7C4

Chthoniobacter flavus El in428 Ammonium transporter B4D7W2
0.69

Methylacidiphilum infernorum _

isolate V4_

Ammonium transporter B3DUX7

0.76

Opitutus ter ae _strain DSM 1 246 / JCM 15787 / PB90-1_ Ammonium transporter B1ZSW9 Chthoniobacter flavus El in428 Ammonium transporter B4CV94

1

0.935

Blastopirel ula marina Ammonium transporter A0A2S8FR08

1

Blastopirel ula marina DSM 3645 Ammonium transporter A3Z L7 Blastopirel ula marina Ammonium transporter A0A2S8GSU9

0.943

Pirel ula staleyi _strain ATCC 2737 / DSM 6068 / ICPB 4128_ Ammonium transporter D2R35

1

Rhodopirel ula sp. SWK7 Ammonium transporter M5SY07

1

0.98

Rhodopirel ula europaea SH398 Ammonium transporter M5S735 Rhodopirel ula europaea 6C Ammonium transporter M2A398

0.747

Rhodopirel ula baltica SH28 Ammonium transporter K5CFE5

0.705

Rhodopirel ula baltica SWK14 Ammonium transporter L7CG1

0.65

Rhodopirel ula baltica WH47 Ammonium transporter F2B2B8

Rhodopirel ula baltica _strain DSM 10527 / NCIMB 1398 / SH1_ Ammonium transporter Q7UYZ3

0.782

0.493
0.82

Acinetobacter bauman i Ammonium transporter A0A9Q8L398

0.842

0.98

Conchiformibius ste dae Ammonium transporter A0A3P2A7W8 1

Neis eria gonor hoeae Ammonium transporter A0A1D3GAE9

Neis eria gonor hoeae _strain ATCC 70 825 / FA 1090_ Ammonium transporter Q5FA36
0.9 7

1

Histophilus somni _strain 129Pt_ Ammonium transporter Q0I4 5 Histophilus somni Ammonium transporter A0A9Q6P4A5

0.81

0.9

Avibacterium paragal inarum Ammonium transporter A0A0F5EZP7 Avibacterium paragal inarum Ammonium transporter A0A37 I7W9

0.9 5

Gal ibacterium anatis 12656/12 Ammonium transporter U1H1F7

0.8

Gal ibacterium anatis 4895 Ammonium transporter A0A0A2ZRL7 Gal ibacterium anatis _strain UMN179_ Ammonium transporter F4H924
0.491

0.976

Alkanindiges il inoisensis Ammonium transporter A0A4Y7XC43

0.945

Acinetobacter sp. AG1 Ammonium transporter A0A0M3AB4

0.91

Acinetobacter haemolyticus ATCC 19194 Ammonium transporter D4XLN6

0.969

Moraxel a catar halis Ammonium transporter A0A7Z0V0A4
0.9 6

Acinetobacter bauman i Ammonium transporter A0A0D5YN10

0.901

0.95

Acinetobacter bauman i 9 063 Ammonium transporter A0A0 9TDB1 Acinetobacter bauman i 625974 Ammonium transporter A0A0 9PYN3

0.734

0.27

Acinetobacter bauman i 149 86 Ammonium transporter A0A836M2B6 Acinetobacter bauman i 834 Ammonium transporter A0A829RI 7

0.543

Acinetobacter bauman i 21072 Ammonium transporter A0A062IU58

0.545

Acinetobacter bauman i Ammonium transporter A0A6F8TLE9

0.9

Acinetobacter bauman i 1462 34 Ammonium transporter A0A9P3CXY2 Acinetobacter bauman i UH5307 Ammonium transporter A0A9P2UCZ8
0.496

Micavibrio aeruginosavorus Ammonium transporter A0A2W5HTC3

0.5

0.781

Persephonel a marina _strain DSM 14350 / EX-H1_ Ammonium transporter C0QTA4
0.7 2

0.958

Hydrogenobacter thermophilus _strain DSM 6534 / IAM 12695 / TK-6_ Ammonium transporter D3DJN2

0.957

0.817

1

Caminibacter mediatlanticus TB-2 Ammonium transporter A6DDR9

Nautil a profundicola _strain ATCC BAA-1463 / DSM 18972 / AmH_ Ammonium transporter B9L7N1
0.38

Desulfurobacterium thermolithotrophum _strain DSM 1 69 / BSA_ Ammonium transporter F0S0G1 Persephonel a marina _strain DSM 14350 / EX-H1_ Ammonium transporter C0QSY0

0.784

0.9

Defer ibacter desulfuricans _strain DSM 14783 / JCM 1 476 / NBRC 101012 / SSM1_ Ammonium transporter D3PAH1

Calditer ivibrio nitroreducens _strain DSM 19672 / NBRC 101217 / Yu37-1_ Ammonium transporter E4TJ50
0.9

Desulfonema limicola Ammonium transporter A0A975GIX9

0.865

Desulfonema magnum Ammonium transporter A0A975BTN0

0.98

Desulfobacula toluolica _strain DSM 7467 / Tol2_ Ammonium transporter K0N2R5 Desulfobacter postgatei 2ac9 Ammonium transporter I5AZQ4
0.797

0.38

Methanosphaera stadtmanae _strain ATCC 43021 / DSM 3091 / JCM 1 832 / MCB-3_ Ammonium transporter Q2NGJ2 Methanobacterium formicicum Ammonium transporter A0A089ZBW5

0.906

Methanothermobacter thermautotrophicus _strain ATCC 29096 / DSM 1053 / JCM 10 4 / NBRC 10 3 0 / Delta H_ Putative ammonium transporter MTH_6 1 O26757

0.985

Methanobrevibacter smithi _strain ATCC 35061 / DSM 861 / OCM 14 / PS_ Ammonium transporter A5UJR1

0.903

Methanobrevibacter smithi DSM 2375 Ammonium transporter B9ACS5

0.59

Methanobrevibacter smithi DSM 2374 Ammonium transporter D2ZRW2

0.87

Methanobrevibacter smithi CAG_186 Ammonium transporter R7PZ71 Methanobrevibacter smithi Ammonium transporter A0A2H4U5R1

0.479

0.91

Paludibacter propionicigenes _strain DSM 17365 / JCM 13257 / WB4_ Ammonium transporter E4T3F7

0.85

Paraprevotel a xylaniphila YIT 1 841 Ammonium transporter F3QPR6 Dysgonomonas gadei ATCC BAA-286 Ammonium transporter F5ISF6

0.493

0.5 2

0.93

0.965

Crinalium epipsammum PCC 93 Ammonium transporter K9W6F3

1

Microcystis aeruginosa NIES-3787 Ammonium transporter A0A6H9GFU4 Microcystis aeruginosa NIES-3804 Ammonium transporter A0A6H9FYW4
0.87

Fischerel a muscicola CCMEE 5323 Ammonium transporter A0A2N6K4Y5
0.963

0.9

Cylindrospermopsis raciborski CENA303 Ammonium transporter A0A1X4G8K1 Cylindrospermopsis raciborski CENA302 Ammonium transporter A0A9Q5QVI9

0.43

Anabaena cylindrica _strain ATCC 2789 / PCC 712 _ Ammonium transporter K9ZG59 Cylindrospermum stagnale PCC 7417 Ammonium transporter K9WT03
0.584

0.83

Fructobacil us fructosus KCTC 354 Ammonium transporter A0A3F3HWV6 Leuconostoc kimchi _strain IMSNU 1 54 / KCTC 2386 / IH25_ Ammonium transporter D5T4N9
0.646

Defer ibacter desulfuricans _strain DSM 14783 / JCM 1 476 / NBRC 101012 / SSM1_ Ammonium transporter D3PE53

0.613

0.38

Geovibrio thiophilus Ammonium transporter A0A410JZR4

Leptospira biflexa serovar Patoc _strain Patoc 1 / ATCC 23582 / Paris_ Ammonium transporter B0SLB7

0.985

Methanosarcina barkeri 3 Ammonium transporter A0A0E3SP24

0.79

Methanosarcina barkeri _strain Fusaro / DSM 804_ Ammonium transporter Q46G49 Methanosarcina barkeri CM1 Ammonium transporter A0A0G3CJ36
0.539

0.713

Al isonel a histaminiformans Ammonium transporter A0A1G5W0P2

0.741

0.853

0.98

Clostridium beijerincki _strain ATCC 51743 / NCIMB 8052_ Ammonium transporter A6M3G0 Clostridium beijerincki _strain ATCC 51743 / NCIMB 8052_ Ammonium transporter A6LV49

0.985

Carnobacterium maltaromaticum LMA28 Ammonium transporter K8EJ02 0.99

Acetobacterium woodi _strain ATCC 29683 / DSM 1030 / JCM 2381 / KCTC 1655 / WB1_ Ammonium transporter H6LKH6 Acetobacterium woodi _strain ATCC 29683 / DSM 1030 / JCM 2381 / KCTC 1655 / WB1_ Ammonium transporter H6LKH9
1

Lactobacil us acidophilus _strain ATCC 700396 / NCK56 / N2 / NCFM_ Ammonium transporter Q5FLR6

0.843

Apilactobacil us kunkeei Ammonium transporter A0A087EQK8 1

Oenococ us oeni Ammonium transporter A0A9Q8YTW8

0.63

Oenococ us oeni ATCC BAA-1 63 Ammonium transporter A0NJ07 Oenococ us oeni AWRIB429 Ammonium transporter D3LBI0

0.62

0.746

Clostridium acetobutylicum _strain ATCC 824 / DSM 792 / JCM 1419 / IAM 19013 / LMG 5710 / NBRC 13948 / NRRL B-527 / VKM B-1787 / 2291 / W_ Ammonium transporter Q97L80

0.745

1

Listeria welshimeri serovar 6b _strain ATCC 35897 / DSM 20650 / CCUG 15529 / CIP 8149 / NCTC 1 857 / SLCC 5334 / V8_ Ammonium transporter A0AIW5 Listeria innocua serovar 6a _strain ATCC BAA-680 / CLIP 1 262_ Ammonium transporter Q92BJ7

1

Bacil us cereus _strain ATCC 14579 / DSM 31 / CCUG 7414 / JCM 2152 / NBRC 15305 / NCIMB 9373 / NCTC 2599 / NRRL B-371 _ Ammonium transporter Q81GN5

0.92

Bacil us thuringiensis subsp. konkukian _strain 97-27_ Ammonium transporter Q6HM26 Bacil us cereus _strain ZK / E33L_ Ammonium transporter Q63EK5
0.71

Clostridium beijerincki _strain ATCC 51743 / NCIMB 8052_ Ammonium transporter A6LRS5
0.718

0.569

0.737

Carboxydothermus hydrogenoformans _strain ATCC BAA-161 / DSM 6008 / Z-2901_ Ammonium transporter Q3AFY9

0.97

Desulfosporosinus acidiphilus _strain DSM 22704 / JCM 16185 / SJ4_ Ammonium transporter I4DBY0

0.92

Desulfosporosinus meridiei _strain ATCC BAA-275 / DSM 13257 / KCTC 12902 / NCIMB 13706 / S10_ Ammonium transporter J7J5E7 Desulfosporosinus orientis _strain ATCC 19365 / DSM 765 / NCIMB 8382 / VKM B-1628 / Singapore I_ Ammonium transporter G7WCB6

0.753

Desulfitobacterium hafniense _strain Y51_ Ammonium transporter Q24PJ2 1

Anaerostipes hadrus Ammonium transporter A0A1Q2C7H7

0.52

Anaerostipes hadrus ATCC 29173 _ JCM 17467 Ammonium transporter L1QB95 Anaerostipes hadrus Ammonium transporter A0A174SD65

0.766

0.955

Dehalobacter restrictus Ammonium transporter A0A857DMN9
0.84

Dehalobacter restrictus Ammonium transporter A0A857DDJ7
0.985

Desulfosporosinus acidiphilus _strain DSM 22704 / JCM 16185 / SJ4_ Ammonium transporter I4D1H5

0.84

Desulfosporosinus meridiei _strain ATCC BAA-275 / DSM 13257 / KCTC 12902 / NCIMB 13706 / S10_ Ammonium transporter J7ITF6 Desulfosporosinus orientis _strain ATCC 19365 / DSM 765 / NCIMB 8382 / VKM B-1628 / Singapore I_ Ammonium transporter G7WA28

0.906

Cetobacterium somerae ATCC BAA-474 Ammonium transporter U7VB41
0.937

0.678

0.634

0.995

Empedobacter brevis NBRC 14943 _ ATCC 43319 Ammonium transporter A0A511NGC7

1

Elizabethkingia anophelis NUHP1 Ammonium transporter A0A077EFU9 Elizabethkingia anophelis Ammonium transporter A0A494JB70
0.385

Coprobacter fastidiosus NSB1 _ JCM 33896 Ammonium transporter A0A495WJM6

0.33

Owenweeksia hongkongensis _strain DSM 17368 / CIP 108786 / JCM 12287 / NRRL B-23963 / UST20020801_ Ammonium transporter G8R7U9 Croceibacter atlanticus _strain ATCC BAA-628 / HTCC2559 / KCTC 12090_ Ammonium transporter A3U857

0.383

0.62

Prolixibacter bellariivorans Ammonium transporter A0A5M4ATH9 Draconibacterium orientale Ammonium transporter A0A1I0BYN1

0.115

Haliscomenobacter hydros is _strain ATCC 27775 / DSM 1100 / LMG 10767 / O_ Ammonium transporter F4KY49
0.13

Myxococ us xanthus _strain DK1622_ Ammonium transporter Q1CZ61 Chloroherpeton thalas ium _strain ATCC 35110 / GB-78_ Ammonium transporter B3QVV5
0.585

0.877

Flavobacterium sp. FPG59 Ammonium transporter A0A243SC06

0.955

Niabella soli DSM 19437 Ammonium transporter W0F0I7
0.61

Mucilaginibacter paludis DSM 18603 Ammonium transporter H1YCC3 Mucilaginibacter paludis DSM 18603 Ammonium transporter H1Y7T6
0.701

0.693

0.95

Niastella koreensis _strain DSM 17620 / KACC 11465 / NBRC 106392 / GR20-10_ Ammonium transporter G8TK77 Arcticibacter svalbardensis MN12-7 Ammonium transporter R9GM11

1

Chitinophaga pinensis Ammonium transporter A0A5C6LWQ1

Chitinophaga pinensis _strain ATCC 43595 / DSM 2588 / LMG 13176 / NBRC 15968 / NCIMB 11800 / UQM 2034_ Ammonium transporter A0A979GQ79

0.43

Runella slithyformis _strain ATCC 29530 / DSM 19594 / LMG 11500 / NCIMB 11436 / LSU 4_ Ammonium transporter A0A7U4E420

0.35

Leadbet erella byssophila _strain DSM 17132 / JCM 16389 / KACC 11308 / NBRC 106382 / 4M15_ Ammonium transporter E4RTK5

0.5

Fibrella aestuarina BUZ 2 Ammonium transporter I0KDC6 Fibrisoma limi BUZ 3 Ammonium transporter I2GD40

0.873

0.566

Parvibaculum lavamentivorans _strain DS-1 / DSM 13023 / NCIMB 13966_ Ammonium transporter A7HSP2

0 5. 64

0.774

0.99

Chromobacterium violaceum Ammonium transporter A0A202B535

Chromobacterium violaceum _strain ATCC 12472 / DSM 30191 / JCM 1249 / NBRC 12614 / NCIMB 9131 / NCTC 9757_ Ammonium transporter Q7NQY2

1

Burkholderia cenocepacia Ammonium transporter A0A088UA43

0.948

Burkholderia cenocepacia Ammonium transporter A0A6J5IRX5

0.858

Burkholderia cenocepacia Ammonium transporter A0A3S9N9F3

0.857

0.75

Burkholderia orbicola _strain AU 1054_ Ammonium transporter A0A0H2XSU4 Burkholderia orbicola _strain MC0-3_ Ammonium transporter B1JZ69

0.55

Burkholderia cenocepacia Ammonium transporter A0A071MMV1 Burkholderia cenocepacia Ammonium transporter A0A1V2XMU9
0.519

0.577

0.899

1

Roseomonas mucosa Ammonium transporter A0A4Y1MSE7 Roseomonas mucosa Ammonium transporter A0A1S8D478

0.96

Granulibacter bethesdensis _strain ATCC BAA-1260 / CGDNIH1_ Ammonium transporter Q0BQ69

0 9. 9

0.953

Kozakia baliensis Ammonium transporter A0A1D8UWJ6

1

Gluconobacter oxydans Ammonium transporter A0A4R4A245

0.8

0.64

Gluconobacter oxydans DSM 3504 Ammonium transporter A0A067Z3S3

Gluconobacter potus Ammonium transporter A0A149QSR6

0.51

Gluconobacter oxydans Ammonium transporter A0A149UA34

0 1. 5

Gluconobacter oxydans Ammonium transporter A0A149RYW7

G ul conobac et r oxydans NBRC 3293 Ammon ui m transpor et r A0A829X098
0.898

0 9. 9

G ul conace ot bac et r d ai zotroph ci us _stra ni ATCC 49037 / DSM 5601 / CCUG 37298 / C PI 103539 / LMG 7603 / PA 5l _ Ammon ui m transpor et r A9HMD6

G ul conace ot bac et r d ai zotroph ci us Ammon ui m transpor et r Q8GQS1

1

0.71

1

Ace ot bac et r trop ci al si Ammon ui m transpor et r A0A149TSA9

0.86

Acetobacter tropicalis Ammonium transporter A0A291PMC6

Acetobacter senegalensis Ammonium transporter A0A149U2G9

1

Acetobacter pasteurianus NBRC 3188 Ammonium transporter A0A401WUW1

0.725

Acetobacter pasteurianus NBRC 3278 Ammonium transporter A0A401X433

0.48

Acetobacter ascendens Ammonium transporter A0A1Y0UZX2

Acetobacter pomorum Ammonium transporter A0A346DY49
0.995

Acetobacter cerevisiae Ammonium transporter A0A149UWF2

0.84

Acetobacter malorum Ammonium transporter A0A149UP06

0.475

Acetobacter malorum Ammonium transporter A0A149V596

0.18

Acetobacter malorum Ammonium transporter A0A149RQX0 Acetobacter cerevisiae Ammonium transporter A0A149QAW7

0.518

0.235 Formivibrio citricus Ammonium transporter A0A1I4ZIZ6 0.22 Paramagnetospirillum magneticum _strain ATCC 700264 / AMB-1_ Ammonium transporter Q2WBF7 Rhodospirillum rubrum Ammonium transporter Q9AM70

0.624

Acidiphilium multivorum _strain DSM 11245 / JCM 8867 / NBRC 100883 / AIU 301_ Ammonium transporter F0J6E8 0.67 0.975 Microvirgula aerodenitrificans Ammonium transporter A0A2S0P9T4 1 Laribacter hongkongensis _strain HLHK9_ Ammonium transporter C1D4X0 Laribacter hongkongensis Ammonium transporter A0A248LMH8 0.788

Parasut erella excrementihominis YIT 11859 Ammonium transporter F3QNA7 0.898 Leptothrix cholodnii _strain ATCC 51168 / LMG 8142 / SP-6_ Ammonium transporter B1Y0E5 0.998 Polaromonas naphthalenivorans _strain CJ2_ Ammonium transporter A1VIM5 0.91 Giesbergeria anulus Ammonium transporter A0A1H9M798 0.995 Delftia acidovorans _strain DSM 14801 / SPH-1_ Ammonium transporter A9BTK1 0.61 Delftia acidovorans Ammonium transporter A0A7T2W1B3 Delftia acidovorans CCUG 274B Ammonium transporter A0A7U9HH57

0.357

0.11

Mariprofundus fer ooxydans PV-1 Ammonium transporter Q0F2M7 Acidiphilium multivorum _strain DSM 11245 / JCM 8867 / NBRC 100883 / AIU 301_ Ammonium transporter F0J3F9
0.368

0.444

0.72

0.68

Methylocella silvestris _strain DSM 15510 / CIP 108128 / LMG 27833 / NCIMB 13906 / BL2_ Ammonium transporter B8EMQ8 Nitrobacter hamburgensis _strain DSM 10229 / NCIMB 13809 / X14_ Ammonium transporter Q1QS12 1 Acidithiobacillus fer ivorans SS3 Ammonium transporter G0JRH4 0.79 Acidithiobacillus fer ivorans Ammonium transporter A0A1B9BUZ8 Acidithiobacillus fer ivorans Ammonium transporter A0A060URB8 0.543 Bauldia litoralis Ammonium transporter A0A1G6ABL9 0.76 Enhydrobacter aerosac us Ammonium transporter A0A1T4SF40 1 Inquilinus limosus Ammonium transporter A0A211ZGL5 Inquilinus limosus MP06 Ammonium transporter A0A0A0D6K5

0.415

0.594

0.955 1

Guillardia theta _strain CCMP2712_ Ammonium transporter L1IYW7 Guillardia theta Ammonium transporter A0A7S4NPK3 0.985 Guillardia theta _strain CCMP2712_ Ammonium transporter L1J6I4 0.75 Guillardia theta _strain CCMP2712_ Ammonium transporter L1IUV5 Guil ardia theta _strain CCMP2712_ Ammonium transporter L1IHZ8 0.828 0.96 Brucel a abortus biovar 1 _strain 9-941_ Ammonium transporter Q57AZ9 Mesorhizobium australicum _strain HAMBI 3006 / LMG 24608 / WSM2073_ Ammonium transporter L0KGR8 0.961 0.92 1 Astic acaulis excentricus Ammonium transporter A0A3G9G4Q4 Astic acaulis excentricus _strain ATCC 15261 / DSM 4724 / KCTC 12464 / NCIMB 9791 / VKM B-1370 / CB 48_ Ammonium transporter E8RSE4 0.955 Brevundimonas subvibrioides Ammonium transporter A0A258HS77 1 Brevundimonas subvibrioides _strain ATCC 15264 / DSM 4735 / LMG 14903 / NBRC 16000 / CB 81_ Ammonium transporter D9QPC8 Brevundimonas subvibrioides Ammonium transporter A0A258FS85 0.935 1 Astic acaulis excentricus Ammonium transporter A0A3G9G2C9 Astic acaulis excentricus _strain ATCC 15261 / DSM 4724 / KCTC 12464 / NCIMB 9791 / VKM B-1370 / CB 48_ Ammonium transporter E8RLT5 0.975 Brevundimonas subvibrioides Ammonium transporter A0A258FTB0 0.9 Brevundimonas subvibrioides _strain ATCC 15264 / DSM 4735 / LMG 14903 / NBRC 16000 / CB 81_ Ammonium transporter D9QPE5 Brevundimonas subvibrioides Ammonium transporter A0A258HPP1 0.586 0.743 Enhydrobacter aerosac us Ammonium transporter A0A1T4SFK4 0.81 Nitrospira defluvi Ammonium transporter D8PD01 1 Rhodopseudomonas palustris _strain BisA53_ Ammonium transporter Q07UM1 0.763 0.98 Rhodopseudomonas palustris _strain ATCC BAA-98 / CGA009_ Ammonium transporter Q6ND36 Rhodopseudomonas palustris Ammonium transporter A0A323UDG5 0.34 Rhodopseudomonas palustris _strain BisB18_ Ammonium transporter Q21CP1 Rhodopseudomonas palustris Ammonium transporter A0A0D7F328 0.722 0.769 1 Micromonas pusil a Ammonium transporter _Fragment_ A0A7S0D560 Micromonas pusil a _strain CCMP1545_ Ammonium transporter C1MX19 0.917 0.76 Maritalea myrionectae Ammonium transporter A0A2R4MBA2 Coraliomargarita akajimensis _strain DSM 45221 / IAM 1541 / JCM 23193 / KCTC 12865 / 04OKA010-24_ Ammonium transporter D5EP71 0.978 0.995 Rhodobacter capsulatus _strain ATCC BAA-309 / NBRC 16581 / SB1003_ Ammonium transporter D5AU98 0.66 Rhodobacter capsulatus Ammonium transporter A0A1G7H8X3 Rhodobacter capsulatus Ammonium transporter A0A4U1JNL6 0.925 Octadecabacter arcticus 238 Ammonium transporter M9RJS5 0.52 Actibacterium mucosum KCTC 23349 Ammonium transporter A0A037ZJY5 Phaeobacter gal aeciensis Ammonium transporter A0A366XFS8 0.732 0.99 Methylococ us capsulatus _strain ATCC 33009 / NCIMB 1 32 / Bath_ Ammonium transporter Q60C43 1 Methylococ us capsulatus _strain ATCC 33009 / NCIMB 1 32 / Bath_ Ammonium transporter Q60BI2 Methylococ us capsulatus Ammonium transporter A0AA35XZV0 0.858 0.993 Methyloglobulus morosus KoM1 Ammonium transporter V5E0F3 0.995 Methylobacter tundripaludum Ammonium transporter A0A2S6GKN5 0.9 Methylobacter tundripaludum _strain ATCC BAA-1 95 / DSM 17260 / SV96_ Ammonium transporter G3IYW0 Methylobacter tundripaludum Ammonium transporter A0A2S6H6B7 0.68 0.36 Alcanivorax borkumensis _strain ATCC 70 651 / DSM 1 573 / NCIMB 13689 / SK2_ Ammonium transporter Q0VQH6 Cel vibrio japonicus _strain Ueda107_ Ammonium transporter B3PIH2 0.792 Sac harophagus degradans _strain 2-40 / ATCC 43961 / DSM 17024_ Ammonium transporter Q21J04 0.79 Catenovulum agarivorans DS-2 Ammonium transporter W7QGE0 0.705 Ruthia magnifica subsp. Calyptogena magnifica Ammonium transporter A1AVV2 1 Colwel ia psychrerythraea Ammonium transporter A0A1Y5E3E3 Colwel ia psychrerythraea _strain 34H / ATCC BAA-681_ Ammonium transporter Q47X17 0.91 0.61 Nitrococ us mobilis Nb-231 Ammonium transporter A4BMY5 0.98 Marichromatium purpuratum 984 Ammonium transporter W0DZS7 Al ochromatium vinosum _strain ATCC 1789 / DSM 180 / NBRC 103801 / NCIMB 104 1 / D_ Ammonium transporter D3RMV7 0.915 0.715 0.705 Hydrocarboniphaga ef usa AP103 Ammonium transporter I8TC71 0.907 Halotalea alkalilenta Ammonium transporter A0A172YCA7 0.9 5 Frateuria aurantia _strain ATCC 3 424 / DSM 62 0 / KCTC 27 / LMG 15 8 / NBRC 3245 / NCIMB 13 70_ Ammonium transporter H8L2Q4 0.9 Dyel a japonica A8 Ammonium transporter A0A075JZ35 Dyel a japonica DSM 16301 Ammonium transporter A0A0G9HEC5 0.864 0.29 Arenimonas oryziter ae DSM 21050 _ YC6267 Ammonium transporter A0A091AXI3 Arenimonas donghaensis DSM 18148 _ HO3-R19 Ammonium transporter A0A087MGU4 0.865 1 0.95 Pseudomonas aeruginosa Ammonium transporter A0A2R3IMJ8 Pseudomonas aeruginosa _strain PA7_ Ammonium transporter A6VE60 1 Pseudomonas aeruginosa Ammonium transporter A0A7M3B3Q4 Pseudomonas aeruginosa _strain UCBPP-PA14_ Ammonium transporter A0A0H2ZKS0 0.635 Azotobacter vinelandi _strain DJ / ATCC BAA-1303_ Ammonium transporter C1DJ63 0.7 Pseudomonas helmanticensis Ammonium transporter A0A4R7VEK0 Pseudomonas berkeleyensis Ammonium transporter A0A7G5DPV0 0.952 Pseudomonas aeruginosa Ammonium transporter A0A3S0KID8 0.95 Hydrocarboniphaga ef usa AP103 Ammonium transporter I7ZHW3

0.95

0.9 2

Salinarimonas soli Ammonium transporter A0A5B2V3B9
0.989

0.915

0.842

1 Neorhizobium galegae bv. of icinalis Ammonium transporter A0A0T7FAP8 Neorhizobium galegae bv. of icinalis Ammonium transporter A0A0T7GZ59

0.86

Rhizobium leguminosarum bv. viciae Ammonium transporter A0A8G2IZM3

0.94 Neorhizobium galegae bv. of icinalis bv. of icinalis str. HAMBI 1 41 Ammonium transporter A0A068TC78 0.64 Neorhizobium galegae bv. orientalis str. HAMBI 540 Ammonium transporter A0A068SVR1 Neorhizobium galegae Ammonium transporter A0A6A1TYJ4 0.813 0.795 1 Caulobacter segnis _strain ATCC 21756 / DSM 7131 / JCM 7823 / NBRC 15250 / LMG 17158 / TK0 59_ Ammonium transporter D5VJ32 Caulobacter segnis Ammonium transporter A0A2W5VAZ2 0.985 Sphingomonas sp. Ag1 Ammonium transporter A0A0M2LU39 0.95 Erythrobacter litoralis _strain HTCC2594_ Ammonium transporter Q2N625 Erythrobacter litoralis Ammonium transporter A0A074ME37 0.738 0.917 0.62 Aurantimonas manganoxydans _strain ATCC BAA-12 9 / DSM 21871 / SI85-9A1_ Ammonium transporter Q1YDY7 Fulvimarina pelagi HTCC2506 Ammonium transporter Q0G0A4 0.8 Haematobacter mis ouriensis Ammonium transporter A0A212AV03 0.9 Aquamicrobium defluvi Ammonium transporter A0A01 U1K7 Pseudaminobacter salicylatoxidans Ammonium transporter A0A316BQ38 0.638 1 Rhodobacter capsulatus Ammonium transporter A0A1G7DX68 Rhodobacter capsulatus _strain ATCC BAA-309 / NBRC 16581 / SB10 3_ Ammonium transporter D5ASL9 0.813 Rubel imicrobium thermophilum DSM 16 84 Ammonium transporter S9QZ09 1 Neorhizobium sp. JUb45 Ammonium transporter A0A4R3A5P7 0.97 Ensifer adhaerens Ammonium transporter A0A9Q9D9I3 Ensifer adhaerens Ammonium transporter A0A0L8BVU4 0.891 0.18 Rhodobacter capsulatus Ammonium transporter A0A4U1K1W1 Rhodomicrobium van ieli _strain ATCC 1710 / ATH 3.1.1 / DSM 162 / LMG 429 _ Ammonium transporter E3I393 0.924 0.785 Azorhizobium caulinodans Ammonium transporter O6 395 0.897 Methylosinus sporium Ammonium transporter A0A2U1SS17 0.785 Beijerinckia indica subsp. indica _strain ATCC 9039 / DSM 1715 / NCIMB 8712_ Ammonium transporter B2IJT1 1 Methylocel a silvestris Ammonium transporter A0A2J7TJ3 Methylocel a silvestris _strain DSM 15 10 / CIP 108128 / LMG 2783 / NCIMB 13906 / BL2_ Ammonium transporter B8ENL9 0.959 0.9 3 0.82 Bradyrhizobium japonicum Ammonium transporter A0A1Y2JH43 Bradyrhizobium japonicum Ammonium transporter A0A1L3F1X5 0.63 Bradyrhizobium japonicum Ammonium transporter A0A0A3XTJ3 1 Bradyrhizobium diazoef iciens _strain JCM 1083 / BCRC 13528 / IAM 13628 / NBRC 14792 / USDA 1 0_ Ammonium transporter Q89WR5 Bradyrhizobium diazoef iciens SEMIA 5080 Ammonium transporter A0A837C6X5 0.981 0.56 Rhodopseudomonas palustris _strain HaA2_ Ammonium transporter Q2J376 Rhodopseudomonas palustris Ammonium transporter A0A418VIJ8 0.9 Rhodopseudomonas palustris _strain BisB5_ Ammonium transporter Q13E03 0.89 Rhodopseudomonas palustris _strain DX-1_ Ammonium transporter E6VIW4 0.743 0.5 Rhodopseudomonas palustris _strain BisA53_ Ammonium transporter Q07UM3 Rhodopseudomonas palustris _strain ATCC BAA-98 / CGA0 9_ Ammonium transporter Q6ND34 0.67 Rhodopseudomonas palustris Ammonium transporter A0A0D7F2Y9 Rhodopseudomonas palustris Ammonium transporter A0A323UE08 0.927 0.497 0.97 Pseudomonas berkeleyensis Ammonium transporter A0A7G5DK18 Leptothrix cholodni _strain ATCC 51 68 / LMG 8142 / SP-6_ Ammonium transporter B1Y6V4 0.795 Prosthecomicrobium hirschi Ammonium transporter A0A0P6VSR8 0.98 Paracoc us denitrificans _strain Pd 12 _ Ammonium transporter A1B3N0 Paracoc us denitrificans Ammonium transporter A0A53 I6R2 0.929 0.851 Burkholderia cenocepacia Ammonium transporter A0A071M4L5 0.893 Methylobacil us flagel atus _strain KT / ATCC 51484 / DSM 6875_ Ammonium transporter Q1H483 0.978 1 Methylococ us capsulatus Ammonium transporter A0A 35XVN2 Methylococ us capsulatus _strain ATCC 3 0 9 / NCIMB 1 32 / Bath_ Ammonium transporter Q605Y6 0.97 Methyloglobulus morosus KoM1 Ammonium transporter V5BYZ7 0.9 5 Methylobacter tundripaludum Ammonium transporter A0A2S6GKF6 0.9 Methylobacter tundripaludum _strain ATCC BA -1 95 / DSM 17260 / SV96

_ Ammonium transporter G3IYW2

Methylobacter tundripaludum Ammonium transporter A0A2S6H643

0.938

0.9 5 Acetobacter ascendens Ammonium transporter A0A1Y0V4A5 0.4 Acetobacter pasteurianus NBRC 318

Ammonium transporter A0A401WXG0
Acetobacter pasteurianus Ammonium transporter A0A1A0DHK0

0.949

0.7 4 0.605

Methyloversatilis universalis

_strain ATCC BA -1314 / DSM 25237 / JCM 13912 / CCUG 52030 / FAM5

_ Ammonium transporter F5RGM7

0.94

Azonexus hydrophilus Ammonium transporter A0A1R1IDV3 Aromatoleum aromaticum _strain EbN1_ Ammonium transporter Q5P470
0.93

0.9

Lysobacter antibioticus Ammonium transporter A0A0S2FHD7 Lysobacter enzymogenes Ammonium transporter A0A0S2DQB6

0.974

Stenotrophomonas rhizophila Ammonium transporter A0A3N1JZV1

0.98

0.9

Luteimonas chenhongjiani Ammonium transporter A0A290XBB5

0.69

Luteimonas ter ae Ammonium transporter A0A4R5U5V1 Luteimonas sp. 9C Ammonium transporter A0A653N397
0.857 0.9

Luteimonas marina Ammonium transporter A0A5C5TYS5 Luteimonas aestuari Ammonium transporter A0A4V3ALP8 1 Luteimonas sp. J16 Ammonium transporter A0A562BXM4 Luteimonas wenzhouensis Ammonium transporter A0A5C5TUT2 0.913

0.3

Gal ionel a capsifer iformans

_strain ES-2

_ Ammonium transporter D9SI59

Desulfobulbus propionicus

_strain ATCC 3 891 / DSM 2032 / VKM B-1956 / 1pr3

_ Ammonium transporter A0A7U3YMN6

0.907

0.562

0.86

Ramlibacter tataouinensis

_strain ATCC BA -407 / DSM 1465

/ LMG 21543 / T

B310

_ Ammonium transporter F5Y3C1

0.8

Ramlibacter tataouinensis Ammonium transporter A0A127JTU2 1 Hylemonel a gracilis str. Niagara R Ammonium transporter A0A016XCG9 1 Hylemonel a gracilis ATCC 19624 Ammonium transporter F3KUN4 Hylemonel a gracilis Ammonium transporter A0A4P6UIK6 0.574 0.666 0.839 0.97 Polaromonas naphthalenivorans _strain CJ2_ Ammonium transporter A1VIG8 0.45 Rhodoferax fer ireducens Ammonium transporter A0A1W9KSU5 Albidiferax fer ireducens _strain ATCC BAA-621 / DSM 15236 / T1 8

_ Ammonium transporter Q2

1S8

0.51 0.98

Pseudacidovorax intermedius Ammonium transporter A0A370FPU0 Pseudacidovorax intermedius Ammonium transporter A0A147GP06

0.55

Leptothrix cholodni

_strain ATCC 51168 / LMG 8142 / SP-6

_ Ammonium transporter B1Y8C9

0.95

Rubrivivax gelatinosus Ammonium transporter A0A4R2M206

Rubrivivax gelatinosus

_strain NBRC 100245 / IL14

_ Ammonium transporter

I0HKQ8

0.997 0.997 0.73

Alcaligenes xylosoxydans xylosoxydans Ammonium transporter A0A176XNC5

Alcaligenes xylosoxydans xylosoxydans Ammonium transporter A0A0X8NXA9

0.98

Achromobacter xylosoxidans

_strain A8

_ Ammonium transporter

E3HJW8

Alcaligenes xylosoxydans xylosoxydans Ammonium transporter A0A424WAH3

0.833

Bordetel a avium

_strain 197N

_ Ammonium transporter Q2KZK9

1 Bordetel a parapertussis

_strain 12822 / ATCC BAA-587 / NCTC 13253

_ Ammonium transporter Q7W7D3

0.48 Bordetel a bronchiseptica

_strain ATCC BAA-588 / NCTC 13252 / RB50

_ Ammonium transporter A0A0H3LUA3

Bordetel a bronchiseptica 253 Ammonium transporter A0A0C6P439
0.872

Hermini monas arsenicoxydans Ammonium transporter A4G1J2
0.852

1 Alicycliphilus denitrificans

_strain DSM 14773 / CIP 107495 / K601

_ Ammonium transporter

F4GED5

0.92 Alicycliphilus denitrificans Ammonium transporter A0A420KGW4 Alicycliphilus denitrificans Ammonium transporter A0A858ZNJ3
0.861 0.892

Giesbergeria anulus Ammonium transporter A0A1H9RTW8

0.97

0.69

Oxalobacter formigenes OXCC13 Ammonium transporter

C3X831

Oxalobacter vibrioformis Ammonium transporter A0A9E9LXV9

1

Oxalobacter ali formigenes Ammonium transporter A0A9E9L9J1 Oxalobacter paraformigenes Ammonium transporter C3X1J1

0.939 0.883

Comamonas piscis Ammonium transporter A0A7G5EC77
0.99 Delftia acidovorans CCUG 274B Ammonium transporter A0A7U9HJZ8 0.56 Delftia acidovorans _strain DSM 14801 / SPH-1_ Ammonium transporter A9BQZ4 Delftia acidovorans Ammonium transporter A0A7T2S0Z6 0.995 Comamonas testosteroni Ammonium transporter A0A8B4RWH8 0.886 Comamonas testosteroni Ammonium transporter A0A373FQ71 0.802 1 Comamonas testosteroni _strain DSM 14576 / KF-1_ Ammonium transporter

B7X047

Comamonas testosteroni TK102

Ammonium transporter A0A076PL76
0.765

Comamonas testosteroni

Ammonium transporter A0A5A7MAJ9
0.68 Comamonas thiooxydans

Ammonium transporter A0A0E3BR91
Comamonas thiooxydans

Ammonium transporter A0A096EEV9

0.874

0.698 0.583

1

Herbaspiril

um

rubrisubalbicans Os34

Ammonium transporter A0A6M3ZJI1
Herbaspiril

um

seropedicae

_strain SmR1

_ Ammonium transporter

D8ITX4

0.61

Pseudoxanthomonas spadix

_strain BD-a59

_ Ammonium transporter

G7UUY6

1

Col imonas fungivorans

Ammonium transporter A0A127PIQ9
Col imonas fungivorans

_strain Ter331

_ Ammonium transporter

G0AFH4

0.903

Pandoraea

pnomenusa

Ammonium transporter A0A378YR15

1 0.92

Ralstonia

solanacearum

_strain Po82

_ Ammonium transporter

F6G5I9

0.97

Ralstonia

syzygi

R24

Ammonium transporter

G3A431

blood disease bacterium A2-HR MARDI

Ammonium transporter

A0A1U9VH69

0.82

Ralstonia

solanacearum

Ammonium transporter

A0AA92K392

0.72

Ralstonia

solanacearum

Ammonium transporter

A0AA92EGF5

0.99 Ralstonia

pseudosolanacearum

Ammonium transporter

A0A223GIX0

Ralstonia

solanacearum

Ammonium transporter

A0A0S4WFM4

0.941

0.912

0.96

Dechloromonas

aromatica

_strain

RCB

_ Ammonium transporter

Q47K18

Propionivibrio dicarboxylicus

Ammonium transporter

A0A1G7VM92

1

Aeromonas dhakensis

Ammonium transporter

K1K628

0.44 Aeromonas hydrophila

subsp. hydrophila

_strain ATCC 7966 / DSM 30187 / BCRC

13018 / CCUG

14551

/ JCM

1027 / KCTC 2358 / NCIMB 9240 / NCTC 8049

_ Ammonium transporter

A0KPE9

Aeromonas hydrophila

subsp. hydrophila

Ammonium transporter

A0A2R4TRE0

0.977

1

Edwardsiel

a tarda

_strain

FL6-60

_ Ammonium transporter

A0A0H3DR67

1

Edwardsiel

a tarda

Ammonium transporter

A0A2A7U6W1

Edwardsiel a tarda ATCC 23685

Ammonium transporter

D4F313

0.953

0.844

1

Morganel a

morgani Ammonium transporter

A0A433ZQ80

Morganel a

morgani subsp. morgani KT

Ammonium transporter

M1SJL5

0.891

0.95

Budvicia

aquatica

Ammonium

transporter

A0A2C6C0X3

Pragia fontium

Ammonium

transporter

A0A0G3CP57

0.903

0.99

Providencia

alcalifaciens

Ammonium

transporter

A0A4R3NLV5

1

Proteus

mirabilis

Ammonium

transporter

A0A1Z1SZE8

Proteus

mirabilis

Ammonium

transporter

A0A2X2C038

0.942

0.998

Providencia

stuartii Ammonium

transporter

A0A1S1HRJ0

0.922

Providencia

burhodogranariea

DSM

19968

Ammonium

transporter

K8W1E2

0.73

0.43

Providencia

ret

geri Ammonium

transporter

A0A264VQM8

Providencia

sneebia

DSM

19967

Ammonium

transporter

K8VZC6

1

Providencia

stuartii _strain

MRSN

2154

_

Ammonium

transporter

A0A140NHJ5

Providencia

stuartii ATCC

25827

Ammonium

transporter

A0AA86YS65

1 0.975

Gilliamella

apicola

Ammonium

transporter

A0A242NIL1

0.71

Gilliamella

apicola

Ammonium

transporter

A0A1B9JHS5

Gilliamella

apicola

Ammonium

transporter

A0A556SRH6

0.851 0.737

Gilliamella

apicola

Ammonium

transporter

A0A1B9MYL7

0.955

Gilliamella

apicola

Ammonium

transporter

A0A1C0A1D9

0.64

Gilliamella

apicola

Ammonium

transporter

A0A2V4EA49

Gilliamella

apicola

Ammonium

transporter

A0A556RTU3

0.703 0.48

Gilliamella

apicola

Ammonium

transporter

A0A1B9KID3

0.675

Gilliamella

apicola

Ammonium

transporter

A0A1B9JPL5

0.99

Gilliamella

apicola

Ammonium

transporter

A0A1B9K8L2

Gilliamella

apicola

Ammonium

transporter

A0A1B9JZS5

0.976

1

Gilliamella

apicola

Ammonium

transporter

A0A1B9MZ78

Gilliamella

apicola

Ammonium

transporter

A0A1B9MBX6

0.892

Gilliamella

apicola

Ammonium

transporter

A0A1B9M5B8

0.905

Gilliamella

apicola

Ammonium

transporter

A0A1B9LFW3

1 0.85

Gilliamella

apicola

Ammonium

transporter

A0A1B9M1A9

Gilliamella

apicola

Ammonium

transporter

A0A1B9NQ91

0.5

Gilliamella

apicola

Ammonium

transporter

A0A1B9N3H7

Gilliamella

apicola

Ammonium

transporter

A0A657K391

0.698 0.886

Serratia

sp. Ag1

Ammonium

transporter

A0A087L0B9

0.975

0.85

Ewingella

americana ATCC

33852

Ammonium

transporter

A0A085G774

Ewingella

americana

Ammonium

transporter

A0A502GPV4

1

Rahnella

sp. JUb53

Ammonium

transporter

A0A4R2Z2K6

0.8

Rahnella

aquatilis

_strain

ATCC

33071

/

DSM

4594

/

JCM

1683

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NBRC

105701

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NCIMB

13365

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CIP

78.65

_

Ammonium

transporter

H2IZ00

Rahnella

sp. BIGb0236

Ammonium

transporter

A0A4R7GKA0

0.71

5

0.986

Brenneria

salicis ATCC

15712

_

DSM

30166

Ammonium

transporter

A0A366IE34

0.955

Pectobacterium

fontis

Ammonium

transporter

A0A7V8IKS6

0.807

Pectobacterium

carotovorum

subsp. carotovorum

_strain

PC1

_

Ammonium

transporter

C6DB68

0.8

Pectobacterium

carotovorum

subsp. carotovorum

Ammonium

transporter

A0AA40J491

0.9

Pectobacterium

carotovorum

subsp. carotovorum

PCC21

Ammonium

transporter

A0AA86DEY7

Pectobacterium

carotovorum

Ammonium

transporter

A0A41

9AVL2

0.698 0.69

1

Pantoea

ananatis

_strain

LMG

201

03

_

Ammonium

transporter

D4GLQ6

Pantoea

ananatis

_strain

AJ1

3355

_

Ammonium

transporter A0A0H3KTD3

0.955

Erwinia

sp. JUb26

Ammonium

transporter A0A3N1VXY4

1

Erwinia

amylovora

_strain

CFBP1430

_

Ammonium

transporter D4HY35

Erwinia

amylovora

ATCC

BAA-21

58

Ammonium

transporter E5B358

0.696

0.97

Citrobacter rodentium

Ammonium

transporter A0A482PLH3

0.835

Salmonella

enterica

subsp. enterica

serovar Lubbock

Ammonium

transporter A0A6C8LST1

1

Escherichia

coli _strain

K1

2

_

Ammonium

transporter AmtB

P69681

Escherichia

coli O1

57

_H7

Ammoniu

m transporter AmtB

P69680

0.662

Enterobacter sp. Ag1

Am

moniu

m transporter A0A9W5JBA9

0.655 0.71

8

0.95

Leclercia

adecarboxylata

Am

moniu

m transporter A0A855EH99

Leclercia

adecarboxylata

Am

moniu

m transporter A0A51

4VEB8

0.98

Enterobacter cloacae

Am

moniu

m transporter A0A7H8UK07

0.98

Enterobacter cloacae

com

plex

sp. 743-2DZ2F-22B

Am

moniu

m transporter A0A4Q2QWS1 Enterobacter cloacae ECNIH2 Ammonium transporter A0A9W3LFI9
0.638

E

nterobacter cloacae

Am

moniu

m transporter A0A427KNL6

0.642 0.873

E

nterobacter cloacae

Am

moniu

m transporter A0A330GB00

0.788 1 E

nterobacter cloacae

Am

moniu

m transporter A0A4Q2EAG7

E

nterobacter cloacae

Am

moniu

m transporter A0A2T4XZC1 0.532 1 Enterobacter cloacae Ammonium transporter A0A6S5K4J1 Enterobacter cloacae complex sp Ammonium transporter A0A2W0GQV0 0.455 Enterobacter cloacae Ammonium transporter A0A443Y093 0.65 Enterobacter cloacae Ammonium transporter A0A144NLT9 Enterobacter cloacae complex sp. ECNIH7 Ammonium transporter A0A1Z3MS36

0.576 0.818

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porter A0A165B

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uxella

agrestis

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porter A0A381CAY8

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agrestis

ATCC

33320

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porter A0A085GI50

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porter A0A2T2Y0

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FY98 1 Raoultella ornithinolytica Ammonium transporter A0A855

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porter A0A1Y6

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LJ1 0.925 Klebsiella michiganensis _strain ATCC 8724 / DSM 4798 / JCM 20051 / N

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021 Klebsiella oxytoca Ammonium transporter A0A168EVK8

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81 0.505 Hafnia alvei Ammonium transporter A0A377PBY1 0.16 Hafnia alvei FB1 Ammonium transporter A0A0

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91 Hafnia alvei Ammonium transporter A0A377PLP5

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0 1 Musicola paradisiaca _strain Ech703_ Ammonium transporter C

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B1 0.66 Dickeya zeae _strain Ech586_ Ammonium transporter D2BUJ8 Dickeya dadantii _strain 3937_ Ammonium transporter E0SD83 0.7 Photorhabdus luminescens subsp. sonorensis Ammonium transporter A

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